
Local Field Theory

— *EYNTKA* Series —

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1

Local Fields

In this chapter, we shall explore special fields which focus into a single prime of a number field. For a prime $p \in \mathbb{Z}$, the fundamental object corresponding to this prime is the p -adic integers $\mathbb{Z}_p$. Intuitively, solving an equation over $\mathbb{Z}_p$ corresponds to solving an equation over $\mathbb{Z} \bmod p^n$ for each n (which can be thought of as solving the equation over all multiplicities). Each $\mathbb{Z}_p$ shall correspond to a field $\mathbb{Q}_p$, and finite extensions of $\mathbb{Q}_p$, $K_{\mathfrak{p}}/\mathbb{Q}_p$, shall only contain primes that are over p . The galois group of these extensions shall then be isomorphic to the decomposition group. The field extensions $K_{\mathfrak{p}}$ shall be called *local number fields*. Finite extensions $K/\mathbb{Q}$ will be called *global number fields*. A large part of the early sections is dedicated to building up the theory of local fields. This culminates with proving the Local-to-Global Correspondence in section 1.8 which readily translates over many results between global fields and local fields.

In the process of developing the generalization, we shall see that the local fields $\mathbb{Q}_p$ correspond very naturally to all possible distinct absolute values that can be defined on $\mathbb{Q}$. In fact, the collection of all possible absolute values that can be defined on $\mathbb{Q}$ shall create the following completions:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, \mathbb{Q}_{11}, \dots, \mathbb{Q}_{\infty} = \mathbb{R}$$

where there is one distinct completion for each prime, as well as one completion that can in fact be naturally interpreted as the completion at the “prime at infinity”. We shall make this idea precise in chapter 2. The idea of places being an extension of primes shall become important when developing class field theory, as well as linking it to the development of extensions of $\mathbb{F}_p(t)$.

Near the end of this chapter, we shall prove the *Kronecker-Weber Theorem* which shows that all abelian extensions are naturally interpreted as being sub-extensions of a cyclic extension. This result shall show a deep connection between galois groups and primes, and shall to great effect be generalized: in chapter ?? we shall develop this completely and show that for every field K , there always exists a field K_H such that the galois group $\text{Gal}_K(K_H)$ is isomorphic to a class group!

1.1 p -adic Integers and p -adic Numbers

Let us start with giving a detailed description of the “new integers” where we only focus on a single prime. Let R be a commutative ring, $\mathfrak{p} \subseteq R$ a prime. Then $R_{\mathfrak{p}}$ will represent the localization of R at $\mathfrak{p}$ (i.e. at the set $R \setminus \mathfrak{p}$). Recall that the completion of R at a prime P is:

$$\varprojlim_n R/\mathfrak{p}^n R = \widehat{R}$$

where the $\mathfrak{p}$ is usually understood from context on the right hand side. For number theory, we are interested in the case where $R = \mathbb{Z}$. This gives us the localizations $\mathbb{Z}_{(p)}$ and the completions $\widehat{\mathbb{Z}}_p$ which we shall denote $\mathbb{Z}_p$. Elements in the completion can be represented as the formal power series:

$$\sum_i^{\infty} a_i p^i = a_0 + a_1 p + a_2 p^2 + \cdots$$

where $a_i \in \{0, \dots, p-1\}$. More succinctly, as $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$, as a sequence

$$(a_0 \bmod p, a_1 \bmod p^2, \dots)$$

It is not immediately clear that these two are the same: in the first we have multiplication via convolution, while in the second we have point-wise multiplication. The equivalence in these two representation can be given by the following: any element can be found by seeing that when we are given:

$$s = \sum_i^{\infty} a_i p^i$$

it has to satisfy:

$$s_n = \sum_i^{n-1} a_i p^i$$

hence, for any $[r] \in \mathbb{Z}/p^n \mathbb{Z}$, there exists a representative of the form:

$$[r] = [a_0 + a_1 p + \cdots + a_{n-1} p^{n-1}]$$

Such an expansion can be written down for any $r \in \mathbb{Z}_{(p)}$, that is for any a/b where $p \nmid b$. From this, if we consider $\mathbb{Z}_p$ to be the formal power series with addition/multiplication extending naturally. From this:

Proposition 1.1.1: Representation Of p -adic

Let $\mathbb{Z}_p$ represent the formal p -adic integers and $\varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$ the completion definition. Then:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$$

Proof:

Put together all the above assertions.

A motivation for defining the p -adic integers is in the following “universal property”: the solutions to Diophantine equations (elements of $\mathbb{Z}[x_1, \dots, x_n]$) mod p^n for all $n \in \mathbb{N}_{>0}$ equivalent to finding solution to Diophantine equations over the p -adics. Note that if we ask for solutions over some collection mod m , by the Chinese Remainder Theorem (CRT) we may reduce the case to solution over powers of primes.

Theorem 1.1.2: Diophantine Equations in p -adic

Let $f \in \mathbb{Z}[x_1, \dots, x_n]$. Then the congruence:

$$f \equiv 0 \pmod{p^n}$$

for all $n \geq 1$ is solvable if and only if f has solutions in the p -adics.

Proof:

Let's first say f has a solution in the p -adics, meaning we make a choice of elements $(x_1, \dots, x_n)$:

$$(x_1, \dots, x_n) = (x_1^{(\nu)}, \dots, x_n^{(\nu)})_{\nu \in \mathbb{N}} \in \mathbb{Z}_p^n$$

such that $f(x_1, \dots, x_n) = 0$. Then by construction we get:

$$f(x_1^{(\nu)}, \dots, x_n^{(\nu)}) \equiv 0 \pmod{p^\nu}$$

Conversely, assume for each $\nu \in \mathbb{N}$ there is a solution

$$f(x_1^{(\nu)}, \dots, x_n^{(\nu)}) \equiv 0 \pmod{p^\nu}$$

If by chance we have that the collection $(x_i^{(\nu)})_{\nu \in \mathbb{N}} \in \prod_{\nu}^{\infty} \mathbb{Z}/p^\nu \mathbb{Z}$ is already in $\varprojlim_{\nu} \mathbb{Z}/p^\nu \mathbb{Z}$, we're done. However, having solutions to moduli equaiton doesn't immediately guarantee the regularity needed for the limit. Instead, a p -adic number will be constructed using these numbers.

We shall construct the construction for the case of $n = 1$, the general case follows. Let $x_\nu = x_1^{(\nu)}$. Then this defines a sequence (x_ν) in $\mathbb{Z}$. As $\mathbb{Z}/p\mathbb{Z}$ has finitely many elements, there are infinitely many elements of the sequence that are congruent to the same element $y_1 \pmod{p}$. Then if $(x_\nu^{(1)})$ is the subsequence:

$$(x_\nu^{(1)}) \equiv y_1 \pmod{p} \quad \text{hence} \quad F(x_\nu^{(1)}) \equiv 0 \pmod{p}$$

From this, another subsequence $(x_\nu^{(2)}) \subseteq (x_\nu^{(1)})$ can be extracted st

$$(x_\nu^{(2)}) \equiv y_2 \pmod{p^2} \quad \text{hence} \quad F(x_\nu^{(2)}) \equiv 0 \pmod{p^2}$$

Continuing on, we get y_i where

$$y_k \equiv y_{k-1} \pmod{p^{k-1}}$$

Hence, we can define $y = (y_k)_{k \in \mathbb{N}} \in \varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$, which is our desired p -addic integer:

$$F(y_k) \equiv 0 \pmod{p^k} \quad \text{if and only if} \quad F(y) = 0$$

as we sought to show.

Example 1.1.3: p -adic Integers

1. Let $n = 10$. Then $10 = 20_5 = 1010_2 = 101_3 = 10_n$ for $n \geq 10$. For non-prime p , we can still do the n -adic completion $\mathbb{Z}_n$, however we will get zero divisors. For example:

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$

which comes from the direct commuting with completion (see [4]).

2. Let $p = 3$ and consider -1_3 . I claim that -1_3 exists. We need to find $[\cdots a_3 a_2 a_1]_3$ such that

$$0001_3 + -1_3 = 0$$

Then $\cdots 222_3$ will give the answer. since:

$$\cdots 222_3 + \cdots 0001_3 = \cdots 000_3$$

Repeat this for any negative number

3. Let $p = 3$ and consider $1/2$. I claim that $(1/2)_3$ exists. We need to find $[\cdots a_3 a_2 a_1]_3$

$$(1/2)_3 \cdot 2_3 = \cdots 0001_3$$

To construct such a number, first:

$$a_1 \cdot 2 \equiv 1 \pmod{3}$$

hence $a_1 = 2$. We then need:

$$2 \cdot a_2 + 1 \equiv 0 \pmod{3}$$

which gives $a_2 = 1$. Continuing we get:

$$(\cdots 1112_3) \cdot (\cdots 0002)_3 = (\cdots 0001)_3$$

This can be done for any prime $p \neq 2$.

4. Let $a/b \in \mathbb{Q}$. Then if $p \nmid b$ there is no p -adic representation. For example, $1/5$ does not have a 5-adic representation (you would need to multiply $1/5$ by 5, but $0005_5 = 0_5$).
5. Using the second example, find the p -adic representation of $\frac{1}{1-p}$.

By the above example, we see that the elements of $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$ can be represented as by the p -adic's $\mathbb{Z}_p$ and hence there is a natural map:

$$\mathbb{Z}_{(p)} \rightarrow \mathbb{Z}_p$$

Note that this map is not surjective (the right hand side is uncountable, or more concretely the right hand side will have $\sqrt{2}$ when $p \neq 2$). This also comes from the more general result that if R is a ring and $\widehat{R}$ is the completion at $\mathfrak{p}$, then if $u \in R/\mathfrak{p}^n$ is a unit, $u \in R/\mathfrak{p}^{2n}$ is a unit. In particular, if $\mathfrak{p} = \mathfrak{m}$ is a maximal ideal, $\widehat{R}_{\mathfrak{m}}$ is a local ring, and hence $\mathbb{Z}_p$ is a local ring (see [4])

Then as $\mathbb{Z}_p$ has a unique maximal ideal, label it $\mathfrak{m}_{\mathfrak{p}}$, let us now consider the localization of $\mathbb{Z}_p$ at $\mathfrak{m}_{\mathfrak{p}}$. Let's denote it $\mathbb{Q}_p$. The completion will consist of all "Laurent series", that is all elements of the

form:

$$\sum_{-m}^{\infty} a_i p^i$$

for all $m \in \mathbb{Z}$, $0 \leq a_i < p$. This allows us to represent *any* rational number, for example if $a/b \in \mathbb{Q}$, then represent it as:

$$\frac{a}{b} = \frac{g}{h} p^{-m} \quad g, h \in \mathbb{Z} \quad \gcd(gh, p) = 1$$

We can then do the p -adic expansion of $g/h = p^m f$:

$$a_0 + a_1 p + a_2 p^2 + \cdots$$

we then shift everything down by m to get the p -adic expansion of f :

$$a_0 p^{-m} + a_1 p^{-m+1} \cdots + a_{m-1} p^{-1} + a_m + a_{m+1} p + a_{m+2} p^2 + \cdots$$

this is called a p -adic number.

1.1.4 Relation to Complex Analysis Notice the similarity to meromorphic functions in complex analysis (this was where the theory of p -adic integers and numbers where inspired). This will be a general trend, where many of the result for number field have exact parallels in function fields; this shall be made more precise in section 1.2.

Hence, we have the more general embedding:

$$\mathbb{Q} \rightarrow \mathbb{Q}_p = (\mathbb{Z}_p)_{\mathfrak{m}_p}$$

where $\mathbb{Z}$ injects into $\mathbb{Z}_p$ (if a, b have the same p -adic expansion, $a - b$ is divisible by p^n for every n , hence $a = b$ in $\mathbb{Z}$). Given this, identify $\mathbb{Q} \subseteq \mathbb{Q}_p$ and $\mathbb{Z} \subseteq \mathbb{Z}_p$.

Let us now take another perspective on $\mathbb{Q}_p$ that is more analytic. Recall that in [4, chapter 19] we classified all absolute values on $\mathbb{Q}$, namely the p -valuations and the usual absolute value. Recall the p -valuation is defined on $\mathbb{Q}$ as:

$$|a|_p = \left| \frac{b}{c} \right|_p = \left| p^m \frac{b'}{c'} \right|_p = p^{-m} \quad \gcd(b'c', p) = 1$$

Where in the p -adic representation of a , we may think of ν_p as counting how many 0's there are before hitting the first non-zero digit, for example in $625 = 1000_5$ we have:

$$|625|_5 = \frac{1}{5^4}$$

Or more algebraically, we take the maximal n where $a \equiv 0 \pmod{p^n}$, $a \not\equiv 0 \pmod{p^{n+1}}$. Hence, the larger the order of the prime p in a/b , the smaller the valuation (intuitively, the “less” detectable the number is when taking $\pmod{p^n}$). This is perhaps intuitive if we compare p -adic expansion to decimal expansion, in which we have:

$$a_0 + a_1 \left(\frac{1}{10} \right) + a_2 \left(\frac{1}{10} \right)^2 + \cdots$$

with $1 \leq a_0 < 10$. Then the more a_i 's that are zero, the smaller the valuation. The above representation of a number in decimal-expansion is good for visualizing a valuation called the ∞ -valuation, $|\cdot|_\infty$ which is the unique Archimedean valuation on $\mathbb{Q}$ up to equivalence (as all p -valuations induce an ultra-metric, they are all non-Archimedean). In this construction, what is important is the exponent, and not the p . The intuition from this comes from the induced p -adic topology, namely from $|\cdot|_p$ we can define the [ultra]-metric

$$d(x, y) = |x - y|_p$$

from which we can construct $\mathbb{Q}_p$ from $\mathbb{Q}$ by taking it's completion using the above metric. Then choosing another constant c will produce equivalent topologies; what matter's for the definition is the factoring of the p and not the "concrete representation".

To continue creating intuition's, let us motivate the $|\cdot|_\infty$ symbol. This comes from considering $k(t)$ over a finite field k , where the analog of $\mathbb{Z}$ is $k[x]$. In $k[x]$, there are monic irreducible polynomials which are in one-to-one correspondence with the primes $\mathfrak{p}$ of $k[t]$ (recall $k[t]$ is a PID). Hence, define $|\cdot|_{\mathfrak{p}} : k(t) \rightarrow \mathbb{R}$ as:

$$f(t) = \frac{g(t)}{h(t)} = p(t)^m \frac{g'(t)}{h'(t)} \quad (g' h', p) = 1$$

Then define $\nu_{\mathfrak{p}}(f) = m$. Then topologically the choice of what to exponentiate doesn't matter, however an informed choice will give us more information, namely let $q_{\mathfrak{p}} = q^{d_{\mathfrak{p}}}$ where q is any real number > 1 and $d_{\mathfrak{p}}$ is the degree of the residue class field of $\mathfrak{p}$ over k (i.e. of $\kappa(f)$). Then:

$$\nu_{\mathfrak{p}}(f) = m \quad |f|_{\mathfrak{p}} = q_{\mathfrak{p}}^{-\nu_{\mathfrak{p}}(f)}$$

Notice that $\nu_{\mathfrak{p}}(0) = \infty$ and $|0|_{\mathfrak{p}} = 0$, giving us that these are indeed the discrete valuations. When $\mathfrak{p} = (t - a)$, the valuation $\nu_{\mathfrak{p}}(f)$ gives the order of the zero/poles of the function $f(t)$ at $t = a$. Then for the infinite case, we have:

$$\nu_{\infty} : k(t) \rightarrow \mathbb{Z} \cup \{\infty\} \quad \nu_{\infty}(f) = \deg(h) - \deg(g)$$

for $f = g/h \neq 0$ and $g, h \in k[t]$. This can be thought of describing the zero and poles at the point at infinity, i.e. the zeros and pole of the function $f(1/t)$ at the point 0. It is in fact associated to a prime ideal, namely $\mathfrak{p} = (1/t)$ in $k[1/t] \subseteq k(t)$. Hence, when considering the absolute value $|\cdot|$ on $\mathbb{Q}$, we may think of this as being the evaluation at the point at infinity, and hence the $|\cdot|_\infty$ symbol. Notice too how this motivates that the size of $|\cdot|_\infty$ diminishes the more zeros' there are in the decimal expansion and taking the reciprocal of integers. As the concrete value of " t " doesn't matter, we can consider any integer and get an expansion that mimics the properties of $|\cdot|_\infty \rightarrow 0$, for example

$$a_0 + a_1 \left(\frac{1}{2}\right) + a_2 \left(\frac{1}{2}\right)^2 + \dots$$

Finally, for $k(t)$, there is also a 0-valuation given by:

$$|f/g|_0 = \text{ord } g - \text{ord } f$$

where the order represents the first degree of the smallest non-trivial monomial, for example $\text{ord}(ax^3 + bx) = 1$. Then in $\mathbb{Q}$, $|\cdot|_0$ is the trivial valuation.

Coming back to absolute values over $\mathbb{Q}$, as what is important is the exponent and not the choice of constant to be exponentiated, it is often better to define a new functional, namely a discrete valuation, which only captures the exponent information. The reader should recall from [4, chapter 20] the following the function $\nu_p : \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$ where $x + \infty = \infty + x = \infty + \infty = \infty$:

1. $\nu_p(a) = \infty$ if and only if $a = 0$ (this mirrors the idea that p divides 0 infinitely many times)
2. $\nu_p(ab) = \nu_p(a) + \nu_p(b)$
3. $\nu_p(a + b) \geq \min(\nu_p(a), \nu_p(b))$

The last condition can be thought of as the ultra-metric condition¹. From this, we may define p -valuations as

$$|-|_p : \mathbb{Q} \rightarrow \mathbb{R} \quad a \mapsto |a|_p = p^{-\nu_p(a)}$$

and we may choose to change p to any other constant $c > 1$.

Now, with the p -adic metric properly defined on $\mathbb{Q}$, we may take the completion of $\mathbb{Q}$ under this metric by taking all Cauchy-sequences of $\mathbb{Q}$ and taking the quotient of all null-sequences (sequences that approach 0). Note that null-sequences form a maximal ideal in the ring of Cauchy-sequences. If R is the ring of Cauchy-sequences and $\mathfrak{m}$, then:

$$\mathbb{Q}_p \cong R/\mathfrak{m}$$

and hence, is an equivalent definition. The reader may see that $\mathbb{Q}_p$ is complete, and that $\mathbb{Q}_\infty = \mathbb{R}$. $\mathbb{Q}$ is naturally embedded in $\mathbb{Q}_p$ via the constant sequence, and as $\mathbb{Q}$ is dense in $\mathbb{Q}_p$ the p -adic valuation naturally extends, namely:

$$|x|_p := \lim_{n \rightarrow \infty} |x_n|_p \in \mathbb{R}$$

where x_n are the elements in the sequence². To show that the $\nu_p(x)$ still represents the exponent (i.e. is an integer), if $x \in \mathbb{Q}_p$ is taken as the class of Cauchy-sequences (where $x_n \neq 0$) then:

$$\nu_p(x_n) = -\log_p |x|_p$$

either diverges or converges is a Cauchy-sequence in $\mathbb{Z}$ which eventually must be constant for large enough n . Then:

$$\nu_p(x) = \lim_{n \rightarrow \infty} \nu_p(x_n) = \nu_p(x_n) \quad \text{for } n \geq n_0$$

Thus, we come full circle and have:

$$|x|_p = p^{-\nu_p(x)}$$

We now have a collection of complete fields with different absolute values:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, \dots, \mathbb{Q}_\infty = \mathbb{R}$$

From each of these, we may extract $\mathbb{Z}_p$ in the following way:

Proposition 1.1.5: Characterizing p -adic Integers

Considering $\mathbb{Q}_p$ as the completion as given above, show that:

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$$

is a well-defined ring. Then $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$ with respect to $|-|_p$

¹And indeed, we can define discrete valuations to be a generalization of non-Archimedean valuations, see [4, chapter 19]

²Technically, this is mod $\mathfrak{m}$; the reader may want to do some practice by checking this is indeed well-defined

Proof:
exercise.

Using the p -adic valuation, we may characterize the units of $\mathbb{Z}_p$ as:

$$\mathbb{Z}_p^* = \{x \in \mathbb{Z}_p \mid |x|_p = 1\}$$

Using this, we may represent every $x \in \mathbb{Q}_p^*$ as :

$$x = p^m u \quad m \in \mathbb{Z}, u \in \mathbb{Z}_p^*$$

that is, every element of $\mathbb{Q}_p^*$ is some unit times some power of prime. For if:

$$\nu_p(x) = m \in \mathbb{Z} \quad \Rightarrow \quad \nu_p(xp^{-m}) = 0 \quad \Rightarrow \quad |xp^{-m}|_p = 1$$

thus $u = xp^{-m} \in \mathbb{Z}_p^*$. Going back to the localization definition of $\mathbb{Z}_p$, this can be seen by remembering that all other primes are localized, and hence any prime factorization $\mathbb{Z}_p$ will now only have a single prime left (and the above argument shows this holds in the completion). From this, we can characterize the ideals of $\mathbb{Z}_p$:

Proposition 1.1.6: Classifying Ideals Of p -adic Integers

Let $\mathbb{Z}_p$ be the p -adic integers. Then set set of nonzero ideals of $\mathbb{Z}_p$ are the principal ideals:

$$p^n \mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \nu_p(x) \geq n\}$$

with $n \geq 0$. Furthermore:

$$\mathbb{Z}_p/p^n \mathbb{Z}_p \cong \mathbb{Z}/p^n \mathbb{Z}$$

Proof:

The first part is a result of the theory of valuations, and is left as an exercise (or check[4])

To show the isomorphism, we'll show that the map $\mathbb{Z} \rightarrow \mathbb{Z}_p/p^n \mathbb{Z}_p$ where $n \mapsto n \bmod p^n \mathbb{Z}_p$ is surjective. To see this, as $\mathbb{Z}_p$ is the completion of $\mathbb{Z}$ with respect to $|\cdot|_p$, for any $x \in \mathbb{Z}_p$ there exists an $a \in \mathbb{Z}$ such that

$$|x - a|_p \leq \frac{1}{p^n}$$

Thus, $x - a \in p^n \mathbb{Z}_p$ (since $n \in \mathbb{Q}$ is in $\mathbb{Z}_p$ if and only if $|n|_p \leq 1$), i.e. $x \equiv a \bmod p^n \mathbb{Z}_p$. Hence, the map is surjective, the kernel consists of all elements where $a \in \mathbb{Z}$ where $a \in p^n \mathbb{Z}_p$, which is exactly the elements of the form $a = p^n x$ for $x \in \mathbb{Z}$, that is the elements $p^n \mathbb{Z}$, hence:

$$\mathbb{Z}_p/p^n \mathbb{Z}_p \cong \mathbb{Z}/p^n \mathbb{Z}$$

as we sought to show.

Using this, we have that:

Proposition 1.1.7: Next Equivalence Of p -adic Integers

Show that with $\mathbb{Z}_p$ as defined from above we have:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

Proof:

As $\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$, we get a surjective homomorphism:

$$\mathbb{Z}_p \rightarrow \mathbb{Z}/p^n\mathbb{Z}$$

This gives a family of homomorphisms which can be used to construct a homomorphism:

$$\mathbb{Z}_p \rightarrow \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

show this is an isomorphism

Finally, let us characterize p -adic integers in one more way and quickly address a possible misconception of the fields $\mathbb{Q}_p$ with respect to $\mathbb{R}$. For the new characterization:

Proposition 1.1.8: Equivalence Of p -adic Via Power-series Quotient

$$\mathbb{Z}_p \cong \mathbb{Z}[[x]]/(x-p)$$

Proof:

exercise.

A common mistake a reader may do is think that $\mathbb{Q}_p \cong \mathbb{R}$ algebraically. We proved this is not the case in [4], however a concrete example should help.

Example 1.1.9: Cautions for p -adic Numbers

1. Recall that $1/p \notin \mathbb{Z}_p$ (review why if this is fuzzy). As $1/p \in \mathbb{Q}_p$, we cannot use it to construct our counter-example. Instead, we shall show that $\sqrt{p} \notin \mathbb{Q}_p$. To see this, notice that $\sqrt{p}$ by definition satisfies the equation $x^2 - p$. Then certainly it would have to satisfy the equation modulo p^n for every n . Then if $\sqrt{p} \in \mathbb{Q}_p$, it must be the case that $\sqrt{p} \in \mathbb{Z}_p$ by theorem 1.1.2. However, $x^2 \equiv p \pmod{p^2}$ has no solutions, and hence $\sqrt{p} \notin \mathbb{Q}_p$.

On the other hand, $\sqrt{1-p^3} \in \mathbb{Q}_p$, as the equations $x^2 - 1 + p^3$ have a solution mod each p^n (use quadratic reciprocity). However, $\sqrt{1-p^3} \notin \mathbb{R}$ as this would require a negative square-root, a number we know that $\mathbb{R}$ does not contain.

2. By field theory, we know that $\mathbb{Q}_p$ embeds into $\mathbb{C}$. This is an example where we really require the axiom of choice. The embeddings is far away from preserving the topological structure of $\mathbb{Q}_p$ or being represented in $\mathbb{C}$. To see this, first notice that $\pm\sqrt{-1} = \pm i \in \mathbb{Q}_5$. It may seem like we should map i to either $\pm i$. However, note that i is closer to 2 while $-i$ is closer to 3. In fact, i is even closer to 57 while the other is much closer to 68. In fact, without

the axiom of choice, this embedding would not be possible. Algebraically, you may also try to check that $\mathbb{Q}_5$ has no automorphisms that swap i and $-i$ (in fact it has *no* non-trivial automorphisms, mirroring how $\mathbb{R}$ has no non-trivial automorphisms^a).

3. Another interesting distinction between $\mathbb{Q}_p$ and $\mathbb{R}$ is the degree of $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ vs $\overline{\mathbb{R}}/\mathbb{R}$. In the latter, we have $\overline{\mathbb{R}} \cong \mathbb{C}$, and we have a degree two extension. After covering Hensel's lemma, we shall see that $x^n - p$ is irreducible for each n in $\mathbb{Q}_p$. It can be shown that there are finitely many extensions for each degree n , and so $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ is a countable extension.

^asee exercise ref:HERE in [4]

double-check intuition here Finally, we introduce the following generalization of prime decompositions to $\mathbb{Q}^3$:

Proposition 1.1.10: Artin-Whaples Product Formula

Let $0 \neq r \in \mathbb{Q}$. Then

$$\prod_{p, \infty} |r|_p = 1$$

where p varies over all primes number and ∞

Proof:

Any $r \in \mathbb{Q}$ can be factored as :

$$r = \pm \prod_{p \neq \infty} p^{\nu_p(r)}$$

and the $\pm$ is equal to $a/|a|_\infty$, giving us the result.

Note that this result work in for the elements of $k(t)$, namely

$$\prod_{\mathfrak{p}, \infty} |f|_{\mathfrak{p}} = 1$$

According to nlab (<https://ncatlab.org/nlab/show/global+field>) global fields are characterized by this product.

Also, I just had another intuition. Recall that the infinite prime feels like it is generically close to all primes. The infinite product tells us that we can define:

$$|r|_\infty := \prod_p |r|_p^{-1}$$

showing that it is define as the product of all other primes, explaining this generic feeling.

Exercise 1.1.1

1. Let $V \subseteq \overline{k}[x_1, \dots, x_n]$ be a variety so that $\mathcal{I}(V) = (f_1, \dots, f_n)$. Let $V(\mathbb{Q})$ be the rational points of the variety. Show that if $V(\mathbb{Q})$ is non-empty, then $V(\mathbb{Q}_p)$ is non-empty for each prime p , and that $V(\mathbb{R})$ is nonempty.

³this result shall be generalized to more general fields in section 2.3

Note the converse need not be true: it was shown that the equation $3x^3 + 4y^3 + 5z^5 = 0$ has a solution over all $\mathbb{Q}_p$ and $\mathbb{R}$, but no real solution. This is an example of the *Hasse Principle* or *local-to-global principle* not being an iff.

1.1.1 Generalization: DVR

This construction started with the integers, i.e. the “fundamental” ring of integers. We now need to have a way to generalize this notion for rings of integers more generally. Recall that Dedekind domains are locally DVRs (see [4, chapter 22.2]). The intuition is that the localization of a Dedekind domain to the DVR gives information of the multiplicity of a prime. Topologically, we may take advantage of the fact that ideals in $\mathbb{Z}_p$ form a chain:

$$\mathbb{Z}_p \supseteq (p) \supseteq (p)^2 \supseteq \dots$$

More generally, as any discrete valuation ring $\mathcal{O}$ has a chain of ideals, we have:

$$\mathcal{O} \supseteq \mathfrak{p} \supseteq \mathfrak{p}^2 \supseteq \dots$$

Then these ideals form a neighborhood-basis:

Definition 1.1.11: Complete DVR

let $\mathcal{O}$ be a DVR with discrete valuation ν . Then $\hat{\mathcal{O}}$ is its completion with respect to ν .

If ν is a normalized exponential valuation (see [4, section 20.7]) and $|\cdot| = q^{-\nu}$ for $q > 1$, we have:

$$\mathfrak{p}^n = \left\{ x \in K \mid |x| < \frac{1}{q^{n-1}} \right\}$$

and we can define a topology on K^* through the sets:

$$U^{(n)} = 1 + \mathfrak{p}^n = \left\{ x \in K^* \mid |1 - x| < \frac{1}{q^{n-1}} \right\} \quad n > 0$$

so that we have the descending chain:

$$\mathcal{O}^* = U^{(0)} \supseteq U^{(1)} \supseteq U^{(2)} \supseteq \dots$$

Notice these are all subgroups under multiplication (use that the valuation is a homomorphism and that $1 + \mathfrak{p}^n = \left\{ x \in K^* \mid |1 - x|_p < \frac{1}{q^{n-1}} \right\}$ for some choice of q to define the p -absolute value). These subgroups are called the *higher unit groups* and $U^{(1)}$ is called the group of *principal units*. These groups have the following property:

Proposition 1.1.12: Higher Unit Group

For $n \geq 1$,

$$\mathcal{O}^*/U^{(n)} \cong (\mathcal{O}/\mathfrak{p}^n)^* \quad \text{and} \quad U^{(n)}/U^{(n+1)} \cong \mathcal{O}/\mathfrak{p}$$

Proof:

Take the surjection $\mathcal{O}^* \rightarrow (\mathcal{O}/\mathfrak{p}^n)^*$. The kernel then is certainly $U^{(n)}$. Similarly for the 2nd isomorphism, where $U^{(n)} = 1 + \pi^n \mathcal{O}$ which maps $1 + \pi^n a \mapsto a \bmod \mathfrak{p}$.

Naturally, the completion of $\mathcal{O}$ is best characterized as a colimit of $\mathcal{O}/\mathfrak{p}^n$:

Proposition 1.1.13: Characterizing Complete DVR Via Colimit

Let $\mathcal{O}$ be a DVR. Then:

$$\mathcal{O} \cong \varinjlim_n \mathcal{O}/\mathfrak{p}^n \quad \mathcal{O}^* \cong \varinjlim_n \mathcal{O}^*/U^{(n)}$$

showing that each complete valuation ring is naturally characterized as a colimit given by its unique maximal ideal.

Proof:

exercise (Neukirch p.128 if you're stuck)

Completion of Discrete Valuation Ring

Let k be a field with a discrete valuation defined on it. This discrete valuation can be extended to $\widehat{k}$ and denoted $\widehat{\nu}$, the completion of k by setting:

$$\widehat{\nu}(a) = \lim_{n \rightarrow \infty} \nu(a_n)$$

Now, for any arbitrarily small ϵ , there exists an a_n such that $|a - a_n| < \epsilon$. Letting $\epsilon = p^{-N}$ for some large n , we see that at some point it must be that:

$$\widehat{\nu}(a - a_n) \geq \widehat{\nu}(a)$$

and so

$$\nu(a_n) = \widehat{\nu}(a_n - a + a) = \min\{\widehat{\nu}(a_n - a), \widehat{\nu}(a)\} = \widehat{\nu}(a)$$

showing that the valuation extends to the completion:

$$\nu(k^*) = \widehat{\nu}(\widehat{k}^*)$$

Proposition 1.1.14: Valuation Ring Quotient And Completion

Let K be a field with valuation ν . Let $\mathcal{O} \subseteq K$ be the valuation ring of K and $\widehat{\mathcal{O}} \subseteq \widehat{K}$ with completed valuation $\widehat{\nu}$. Then:

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$$

If ν is discrete, then:

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n \cong \mathcal{O}/\mathfrak{p}^n$$

In particular, the size of the residue field, $\#\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}$, is finite.

Proof:

generalize proposition 1.1.6

Using this, the p -adic expansion extends to the completion of any discrete valuation of K (see Neukirch p.127 for more details). The main advantage of this generalization is to show that the power-series expansion also works for function fields and we get fields of formal power series $K((x))$ with the infinite expansion representation.

Exercise 1.1.2

1. Show that $\sqrt{2} \in \mathbb{Q}_p$ if and only if $x^2 - 2$ factor into linear factors mod p .

1.2 Local and Global Fields

An overarching theme in this entire book is to understand the relation between a field (such as a number field) and its localizations. It was already noted in 1.1.4 that function field and number fields share many of the same characteristics when localizing. Let us box these notions to study this further:

Definition 1.2.1: Global And Local Fields

1. A *global field* is a finite extension of either $\mathbb{Q}$ or $\mathbb{F}_p(t)$.
2. A *local field* is a field which is complete with respect to a discrete valuation ν and its residue field is finite.

Note that the valuation on the completion of global fields is discrete and has a finite residue class field. If we don't specify the finite residue field, we shall get some constructions we are not looking for. We should immediately re-characterize local fields to make them easier to work with:

Proposition 1.2.2: Characterizing Local Fields

Let K be a local field. Then either:

1. K is a finite extension of $\mathbb{Q}_p$
2. K is a finite extension of $\mathbb{F}_p((t))$

Some authors will define local fields to be completion of fields with respect to an absolute value to allow for $\mathbb{R}$ to be a local field, but this will not be done here **In particular, Varma does this.**

Proof:

Neukirch p.135

I will for now take this as granted, for in my head local fields can be seen as being these by definition. The point of the abstraction is to unify them and not need to keep taking cases, but I will be okay with that for now.

Hence, the local fields of characteristic $p \neq 0$ are of the form $\mathbb{F}_q((t))$ where $q = p^n$, and local fields of characteristic 0 are finite extensions of $K/\mathbb{Q}_p$. The former will be called *power series fields*, and the latter will be called *p-adic number fields*.

Proposition 1.2.3: Compactness Of Local Fields

Let K be a local field. Then K is locally compact, and its valuation ring $\mathcal{O}$ is compact

Proof:

By 1.1.13, $\mathcal{O} \cong \varprojlim \mathcal{O}/\mathfrak{p}^n$ algebraically and topologically. By 1.1.12, $\mathfrak{p}^\nu/\mathfrak{p}^{\nu+1} \cong \mathcal{O}/\mathfrak{p}$, and since $\mathcal{O}/\mathfrak{p}^n$ are finite, they are finite. As $\mathcal{O}$ is a closed subset of the compact product $\prod_{n=1}^{\infty} \mathcal{O}/\mathfrak{p}^n$, we have that the limit is compact, and hence $\mathcal{O}$ is. Next, for every $a \in K$, $a + \mathcal{O}$ is open and a compact neighborhood, so K is locally compact.

For the topologically minded, here is an interesting result: if K is a locally compact field with respect to a nondiscrete topology, it is isomorphic to euclidean $\mathbb{R}$ or $\mathbb{C}$, or to a local field with the valuation topology. Hence, we can say that we are studying locally compact fields.

Definition 1.2.4: Local Number Field

Let $K_{\mathfrak{p}}/\mathbb{Q}_p$ be a finite extension of $\mathbb{Q}_p$. Then $K_{\mathfrak{p}}$ is called a *local number field*

Notice that the extension has a sub-script $\mathfrak{p}$ attached to it. Each local number field shall be associated to a prime, as theorem 1.4.2 suggests (but we shall prove it in the proceeding sections).

Proposition 1.2.5: Units Of Local Field

Let K be a local field. Then:

$$K^* = (\pi) \times \mu_{q-1} \times U^{(1)}$$

where π is a prime element, $q = \#(\kappa(\mathfrak{p}) = \#(\mathcal{O}/\mathfrak{p})$, and $U^{(1)} = 1 + \mathfrak{p}$ is the group of principal units

Proof:

For any $x \in K^*$, we have that $k = \pi^n u$ for a appropriate prime π , $n \in \mathbb{Z}$, and $u \in \mathcal{O}^*$. As $\mathcal{O}$ contains the $q - 1$ roots of unity μ_{q-1} (by Hensel's lemma, $x^{q-1} - 1$ splits in $\mathcal{O}$). The homomorphism $\mathcal{O}^* \rightarrow \kappa^*$ mapping $u \mapsto u \bmod \mathfrak{p}$ has $U^{(1)}$ as a kernel and the group of unity maps bijectively onto κ^* , hence $\mathcal{O}^* = \mu_{q-1} \times U^{(1)}$, as we sought to show.

Proposition 1.2.6: Log For p-adic Number Fields

Let K be a p -adic number field. Then there exists a unique continuous homomorphism:

$$\log : K^* \rightarrow K$$

where $\log p = 0$ and on principal ideals $(1 + x) \in U^{(1)}$ is given by:

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

Proof:

Neukirch p.137 if you want the proof.

(there is one more lemma on p.137 of Neukirch that gives maps $\mathfrak{p}^n \rightleftharpoons \exp_{\log} U^{(n)}$)

(using these results, we get a more specific decomposition of the group)

(5.7) Proposition. *Let K be a local field and $q = p^f$ the number of elements in the residue class field. Then the following hold.*

(i) *If K has characteristic 0, then one has (both algebraically and topologically)*

$$K^* \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^d,$$

where $a \geq 0$ and $d = [K : \mathbb{Q}_p]$.

(ii) *If K has characteristic p , then one has (both algebraically and topologically)*

$$K^* \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}_p^{\mathbb{N}}.$$

1.2.7 Henselian Fields If you are following Neukirch, this point he takes some time to develop Henselian fields, as it turns out that most of the theory only requires Hensel's lemma (what is covered in the next section), which applies to fields that are more general than Complete fields and local fields. I will omit this for now, as it is not necessary at this stage

The gist is: If R is a DVR, it's henselization is $\widehat{R} \cap K^{\text{sep}}$, where $K = \text{Frac}(R)$. The idea is that Cauchy sequences that converge (in the completion) to the root of a polynomial are *required* to converge, but not every Cauchy sequence needs to converge!

These also show up in algebraic geometry when taking completion at a point of $\text{Spec } R$; we technically add more elements in the global sections than is needed algebraically (the completion will add power series that are not algebraic, ex. possibly e^x), thus we could instead take the *Henselization* instead of the *completion* at a point

1.3 Hensel's Lemma

Hensel's lemma can be thought as the foundational result on why we want to work over local fields. Recall that given a polynomial $f(x) \in \mathbb{Z}[x]$, it is possible that $f(x)$ has a root mod each prime, but f need not have a root. For example recall that $x^4 + 1 \in \mathbb{Z}[x]$ is irreducible (use the fact that $\sqrt{2}$ is irrational in the proof). However, this polynomial is reducible mod every prime (see [4, chapter 11.3]). When working over local fields, it was shown in theorem 1.1.2 that a polynomial has a solution if it has a solution mod each p^n (or equivalently for $\varprojlim_n R/\mathfrak{m}^n$, for each $\mathfrak{m}^n$). Hensel's lemma says that under mild conditions, it suffices to check that a polynomial has a root mod p (resp. mod $\mathfrak{m}$) for it to have a root mod for each p^n (resp $\mathfrak{m}^n$)! In particular, Hensel's lemma give a way of lifting roots.

Let K be a complete field with respect to to some nonarchemedian absolute value $|\cdot|$, and hence has a corresponding valuation. Let $\mathcal{O} \subseteq K$ be the valuation ring with corresponding unique maximal ideal $\mathfrak{p}$ and residue field $\kappa(\mathfrak{p}) = \mathcal{O}/\mathfrak{p}$. For example, $K = \mathbb{Q}_p$ and $\mathcal{O} = \mathbb{Z}_p$ with ideal (p) and $\kappa((p)) = \mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$. Define a *primitive polynomial* $f(x) \in \mathcal{O}[x]$ if

$$f(x) \not\equiv 0 \pmod{\mathfrak{p}}$$

which similarly means

$$|f| = \max\{|a_0|_{\mathfrak{p}}, \dots, |a_n|_{\mathfrak{p}}\} = 1$$

(recall that if $a_i = np^k$ for some $p \in \mathfrak{p}$, $p \notin \mathfrak{p}^2$ then $|a_i|_{\mathfrak{p}} = p^{-k}$). If $\mathcal{O}$ is a Dedekind domain, this is equivalent to saying $\gcd(a_1, \dots, (a_n)) = (1)^4$. Note that it need not be monic. Then factorization mod $\mathfrak{p}$ gives factorization when lifted!

Theorem 1.3.1: Hensel's Lemma

Let K be a complete field with respect to a nonarchemedian absolute value $|\cdot|$, and let $\mathcal{O}$ be the corresponding discrete valuation ring. Let $f(x) \in \mathcal{O}[x]$ be a primitive polynomial. Then if it admits a factorization modulo $\mathfrak{p}$:

$$f(x) \equiv \bar{g}(x)\bar{h}(x) \pmod{\mathfrak{p}}$$

into relatively prime polynomials $\bar{g}, \bar{h} \in \kappa[x]$, then $f(x)$ admits a factorization

$$f(x) = g(x)h(x)$$

for polynomials $g, h \in \mathcal{O}[x]$ such that the degree of at least g matches, and:

$$g(x) \equiv \bar{g}(x) \pmod{\mathfrak{p}} \quad \text{and} \quad h(x) \equiv \bar{h}(x) \pmod{\mathfrak{p}}$$

Proof:

We shall inductively construct g, h . First, let $d = \deg(f)$, $\deg(\bar{g}) = m$, and hence $\deg(\bar{h}) \leq d - m$. Let $g_0, h_0 \in \mathcal{O}[x]$ be polynomial representing $\bar{g}$ and $\bar{h}$ respectively where $\deg(g_0) = \deg(\bar{g})$ and $\deg(h_0) \leq d - m$. As $(\bar{g}, \bar{h}) = 1$ are co-prime, there exists $a, b \in \mathcal{O}[x]$ such that $ag_0 + bh_0 = 1$, hence $ag_0 + bh_0 \equiv 1 \pmod{\mathfrak{p}}$. Now, the trick is to pick among the coefficients of $f - g_0h_0$, $ag_0bh_0 - 1 \in \mathfrak{p}[x]$ the one with minimum value with respect to $|\cdot|$ (convince yourself this polynomials are in $\mathfrak{p}[x]$).

⁴Note that not all rings of integers have this conditions (in [1] rank-2 valuation on $\mathbb{Q}(x, y)$)

Call this element π .

Using this element, we shall construct g, h that will be the factorizations through the following form:

$$\begin{aligned} g &= g_0 + p_1\pi + p_2\pi^2 + \cdots \\ h &= h_0 + q_1\pi + q_2\pi^2 + \cdots \end{aligned}$$

where $p_i, q_i \in \mathcal{O}[x]$ are polynomials of degree $< m$ and $\leq d - m$ respectively. These are indeed elements of $\mathcal{O}[x]$ when constructed as the limit:

$$\begin{aligned} g_{n-1} &= g_0 + p_1\pi + p_2\pi^2 + \cdots + p_{n-1}\pi^{n-1} \\ h_{n-1} &= h_0 + q_1\pi + q_2\pi^2 + \cdots + p_{n-1}\pi^{n-1} \end{aligned}$$

where

$$f \equiv g_{n-1}h_{n-1} \pmod{\pi^n} \tag{1.1}$$

as the term will always group up to form a polynomial of appropriate degree. What's to do is to show that we can find polynomials g_{n-1}, h_{n-1} that satisfy equation (1.1) which by a small generalization of theorem 1.1.2 shall give the desired polynomial.

For $n = 1$, the solution is immediately given by choice of π . Let us now inductively assume that it works for some $n - 1$, and consider:

$$g_n = g_{n-1} + p_n\pi^n \quad h_n = h_{n-1} + q_n\pi^n$$

Then:

$$f - g_{n-1}h_{n-1} \equiv (g_{n-1}q_n + h_{n-1}p_n)\pi^n \pmod{\pi^{n+1}}$$

As we are in an integral domain, inverses are well-defined and hence we may divide by π^n . This gives:

$$g_{n-1}q_n + h_{n-1}p_n \equiv (g_0q_n + h_0p_n) \equiv \pi^{-n}(f - g_{n-1}h_{n-1}) \equiv f_n \pmod{\pi}$$

where $f_n = \pi^{-n}(f - g_{n-1}h_{n-1}) \in \mathcal{O}[x]$. Next, since $g_0a + h_0b \equiv 1 \pmod{\pi}$, we have:

$$g_0af_n + h_0bf_n \equiv f_n \pmod{\pi}$$

If we can take $q_n = af_n$ and $p_n = bf_n$, we'd be done, however the degree's may be too big. Hence, write:

$$b(x)f_n(x) = q(x)g_0(x) + p_n(x)$$

where $\deg(p_n) < \deg(g_0) = m$. Since $g_0 \equiv \bar{g} \pmod{\mathfrak{p}}$ and $\deg(g_0) = \deg(\bar{g})$, the highest coefficient of g_0 is a unit, and so $q(z) \in \mathcal{O}[x]$. Thus, we get:

$$g_0(af_n + h_0q) + h_0p_n \equiv f_n \pmod{\pi}$$

Finally, omitting the coefficient divisible by π in the polynomial af_nh_0q , we get q_n such that $g_0q_n + h_0p_n \equiv f_n \pmod{\pi}$, where $\deg(f_n) \leq d$, $\deg(g_0) = m$, and $\deg(h_0p_n) < (d - m) + m = d$, meaning it has degree less than or to $d - m$, giving the induction step, and completing the proof.

Corollary 1.3.2: Hensel's Lemma – Lifting Roots

Let R be a complete DVR with maximal ideal $\mathfrak{p}$ and residue field $k = R/\mathfrak{p}$. Then if f is a monic polynomial whose reduction to $k[x]$ has a *simple* root $\bar{a} \in k$, then $\bar{a}$ can be lifted to a root of f in R .

Proof:

If $\bar{a} \in k = R/\mathfrak{p}$ is a root, then by Hensel's lemma it lifts to a root in R .

Example 1.3.3: Hensel's Lemma

1. (roots of unity) Consider $x^{p-1} - 1 \in \mathbb{Z}_p[x]$. Over $\mathbb{F}_p \cong \mathbb{Z}_p/p\mathbb{Z}_p$, this polynomial splits completely into linear factors. Hence through repeated use of Hensel's lemma it splits in $\mathbb{Z}_p$. As such, the field $\mathbb{Q}_p$ of p -adic number contains the $(p-1)$ roots of unity!
These along with 0 can even form a system of representatives for the residue class field (the usual representatives are $0, 1, \dots, p-1$).
2. Consider $x^2 + 1$. Then in mod 2, we have that $x^2 + 1 \equiv (x+1)^2 \pmod{2}$, and hence Hensel's lemma cannot lift -1 . By theorem 1.1.2, notice that $x^2 + 1 \pmod{4}$ has no solutions, hence $x^2 + 1$ has no root in $\mathbb{Q}_2$.
3. Hensel's lemma is a sufficient, but not necessary condition. Consider $x^2 - p^2$ over $\mathbb{Q}_p$. Then $x^2 - p^2 \equiv x^2 \pmod{p}$ has multiple roots, hence Hensel's lemma does not apply. However, p of $x^2 - p^2$ over $\mathbb{Q}_p$, namely $\pm p$. The problem can be seen as there being two possible lifting of $0 \pmod{p}$.
4. Show that $x^2 - 98 = x^2 - 2 \cdot 7^2$ in $\mathbb{Q}_7$ has two roots (hint: quadratic reciprocity). However again it does not satisfy the condition of Hensel's lemma.

Corollary 1.3.4: Hensel's Lemma Giving Newtons Method

Let R be a complete DVR, $f \in R[x]$, and $a_0 \in R$ satisfies:

$$|f(a_0)| < |f'(a_0)|^2$$

implying $f'(a_0)$ divides $f(a_0)$. For any $n \geq 0$ define

$$a_{n+1} := a_n - \frac{f(a_n)}{f'(a_n)}$$

Then the sequence (a_n) is well-defined and converges to a unique root $a \in R$ for which

$$|a - a_0| \leq \epsilon := \frac{|f(a_0)|}{|f'(a_0)|^2}$$

Furthermore, $|f(a_n)| < \epsilon^{2n} |f'(a_0)|^2$ for all $n \geq 0$

Proof:

look at Newton's method. (for ex.[5])

Corollary 1.3.5: Hensel-Kürschák Lemma

Let K be a complete field with respect to a non-Archimedean valuation $|\cdot|$. Then for every irreducible polynomial $f(x) = a_0 + \cdots + a_n x^n \in K[x]$ such that $a_0 a_n \neq 0$, we have:

$$|f| = \max\{|a_0|, |a_n|\}$$

In particular, if $a_n = 1$ and $a_0 \in \mathcal{O}$, then $f \in \mathcal{O}[x]$

This is a rather powerful result, as it limits the possible irreducible polynomials over such fields!

Proof:

By multiplying by an appropriate element of K , assume that $f \in \mathcal{O}[x]$ and $|f| = 1$. Let a_r be the first coefficient where $|a_r| = 1$ so that:

$$f(x) \equiv x^r(a_r + a_{r+1}x + \cdots + a_n x^{n-r}) \pmod{\mathfrak{p}}$$

Then $\max\{|a_0|, |a_n|\} < 1$, $0 < r < n$, but then the congruence contradicts Hensel's lemma, as we sought to show.

1.4 Extension of Local Fields

With the definition of the local fields, and the ground work for how to complete any discrete valuation, we now look at how valuations change under field extensions. In this whole section, we shall be consider the following scenario all the time:

Let A be an integral domain, K the field of fractions of A , L a finite extension of K , and B the integral closure of A in L . This setup shall be called the *AKLB setup*.

We shall put some different conditions on A depending on what we need. In number theory, we usually have A be a Dedekind domain.

Recall that each prime $\mathfrak{p} \subseteq A$ where A is a Dedekind domain determines a discrete valuation $\nu_{\mathfrak{p}} : \mathcal{I}_A \rightarrow \mathbb{Z}$ by giving the exponent $n_{\mathfrak{p}}$ from the unique factorization (i.e. $\nu_{\mathfrak{p}}(I) = n$ if and only if $IA_{\mathfrak{p}} = \mathfrak{p}^n A_{\mathfrak{p}}$). Thus for each prime $\mathfrak{p} \subseteq A$, we have the discrete valuation:

$$\nu_{\mathfrak{p}} : K \rightarrow \mathbb{Z} \quad \nu_{\mathfrak{p}}(x) = \nu_{\mathfrak{p}}(xA)$$

and hence an absolute value $|x|_{\mathfrak{p}} := c^{\nu_{\mathfrak{p}}(x)}$ for $0 < c < 1$. If we have *AKLB*, primes $\mathfrak{P}/\mathfrak{p}$ also give rise to a discrete valuation $\nu_{\mathfrak{P}}$ on L , each restricts to a valuation on K , but the valuation need not be equal to $\nu_{\mathfrak{p}}$. Let us start by understanding the relation

Definition 1.4.1: Extending Valuations

Let L/K be a finite separable extension, and let ν, w be discrete valuation on K, L respectively. Then if $w|_K = e\nu$ for some $e \in \mathbb{N}_{>0}$, then w is said to *extend ν with index e*

Theorem 1.4.2: Extension of Valuation

Let $AKLB$, $\mathfrak{p} \subseteq A$ a prime, and $\mathfrak{P}/\mathfrak{p}$ a prime in B lying over $\mathfrak{p}$. Then there is a discrete valuation $\nu_{\mathfrak{P}}$ which extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and every discrete valuation on L that extends $\nu_{\mathfrak{p}}$ arises in this way. Thus, we have a bijection $\mathfrak{P} \rightarrow \nu_{\mathfrak{P}}$ from $\mathfrak{P}/\mathfrak{p}$ to the set of discrete valuation on L that extend $\nu_{\mathfrak{p}}$.

Proof:

Let $\mathfrak{P}/\mathfrak{p}$ so that $\nu_{\mathfrak{P}}(\mathfrak{p}B) = e_{\mathfrak{P}}$ by definition, and if $\mathfrak{p} \neq \mathfrak{p}'$ then $\nu_{\mathfrak{P}}(\mathfrak{p}'B) = 0$ as $\mathfrak{P}$ lies above only one prime. Now if $I = \prod_{\mathfrak{p}'} (\mathfrak{p}')_{n_{\mathfrak{p}'}}$ is a nonzero fractional ideal then:

$$\nu_{\mathfrak{P}}(IB) = \nu_{\mathfrak{P}} \left(\prod_{\mathfrak{p}'} (\mathfrak{p}')_{n_{\mathfrak{p}'}} B \right) = \nu_{\mathfrak{P}}(\mathfrak{p}^{n_{\mathfrak{p}}} B) = \nu_{\mathfrak{P}}(\mathfrak{p}B)n_{\mathfrak{p}} = e_{\mathfrak{P}}n_{\mathfrak{p}} = e_{\mathfrak{P}}\nu_{\mathfrak{p}}(I)$$

Thus, $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$.

Furthermore, if $\mathfrak{P}, \mathfrak{P}'$ are two distinct primes lying over $\mathfrak{p}$, then neither contains each other, and any $x \in \mathfrak{P} - \mathfrak{P}'$ we have

$$\nu_{\mathfrak{P}}(x) > 0 \geq \nu_{\mathfrak{P}'}(x)$$

and so $\nu_{\mathfrak{P}} \neq \nu_{\mathfrak{P}'}$, and so $\mathfrak{P} \mapsto \nu_{\mathfrak{P}}$ is injective.

To show that every extension of valuation arises in this way, let w be a discrete valuation on L that extends $\nu_{\mathfrak{p}}$ with index e . Let

$$W = \{x \in L \mid w(x) \geq 0\}$$

be the associated DVR and let $\mathfrak{m}$ be the associated maximal ideal. Since $w|_K = e\nu_{\mathfrak{p}}$ by definition of w , the discrete valuation on w is nonzero on A , and hence $A \subseteq W$. The elements of A with nonzero valuation are exactly the elements of $\mathfrak{p}$, and hence

$$\mathfrak{p} = \mathfrak{m} \cap A$$

Furthermore, the DVR W is integrally closed in its fraction field L (see [4]), and so $B \subseteq W$ by minimality for B . Now, $\mathfrak{P}$ is prime (since $\mathfrak{m}$ is) and $\mathfrak{p} = \mathfrak{m} \cap A = \mathfrak{P} \cap A$, and so $\mathfrak{P}$ lies over $\mathfrak{p}$. The ring W contains $B_{\mathfrak{P}}$ and is contained in $\text{Frac} B_{\mathfrak{P}} = L$. But now, there is no intermediate rings between DVR and fraction fields (see [4]), and so $W = FB_{\mathfrak{P}}$, and $w = \nu_{\mathfrak{P}}$ with $e = e_{\mathfrak{P}}$, completing the proof.

With this, we return to upgrading our AKLB setup to local fields. We have that that the extension of complete DVR's is "closed". We shall next upgrade the following:

Lemma 1.4.3: Detecting Integral Elements, Complete DVR

Let A be a complete DVR with fraction field K , and L/K be a finite extension of degree n . Then $\alpha \in L$ is integral over A if and only if $N_{L/K}(\alpha) \in A$.

Recall that in the case for number fields, this was *not* a sufficient condition [4, chapter 22.1].

Proof:

Let $f = \sum_i a_i x^i \in K[x]$ be the minimal polynomial of α . If α is integral over A , then $f \in A[x]$, meaning all its coefficients are in A including the constant, which is equal to $N_{L/K}(\alpha)^e$ ($e = [L : K(\alpha)]$). Conversely, if $N_{L/K}(\alpha) = (-1)^n f(0)^e \in A$, then $f(0) \in A$ since $f(0) \in K$ is the roots of $x^e - (-1)^n N_{L/K}(\alpha) \in A[x]$ and A is integrally closed. Thus, the constant coefficient of f lies in A , and as it is monic, so does f , and so by corollary 1.3.5 $f \in A[x]$, showing that α is integral, completing the proof.

Theorem 1.4.4: Extension of DVR's

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Then B is a DVR whose maximal ideal $\mathfrak{P}$ is necessarily the unique prime ideal above $\mathfrak{p}$.

Proof:

Let's first show that the number of primes lying over $\mathfrak{p}$ is 1. There is at least one $\mathfrak{P} \subseteq B$ lying over $\mathfrak{p}$ as the factorization of $\mathfrak{p}B \subseteq B$ is non-trivial by theorem 1.4.2. Now, for the sake of contradiction say that there were $\mathfrak{P}, \mathfrak{P}'$ lying over $\mathfrak{p}$ where $\mathfrak{P} \neq \mathfrak{P}'$. Then choose $b \in \mathfrak{P} - \mathfrak{P}'$ and consider $A[b] \subseteq B$. The ideals $\mathfrak{P} \cap A[b]$ and $\mathfrak{P}' \cap A[b]$ are distinct prime ideals of $A[b]$ containing $\mathfrak{p}A[b]$. Furthermore, both are maximal since they are nonzero and $\dim A[b] = \dim(A) = 1$ (as $A[b]$ is integral over A , and therefore has the same dimension). The quotient ring $A[b]/\mathfrak{p}A[b]$ thus has *at least* two maximal ideals. Now let $f \in A[x]$ be the minimal polynomial of b over K and let $\bar{f} \in (A/\mathfrak{p})[x]$ be its reduction to the residue field $A/\mathfrak{p}$. Then we have:

$$\frac{(A/\mathfrak{p})[x]}{(\bar{f})} \cong \frac{A[x]}{(\mathfrak{p}, f)} \cong \frac{A[b]}{\mathfrak{p}A[b]}$$

Hence, the ring $(A/\mathfrak{p})[x]/(\bar{f})$ also has at least two maximal ideals, implying that $\bar{f}$ is divisible by two distinct irreducible polynomials (as $(A/\mathfrak{p})[x]$ is a PID). Thus, we get that $\bar{f} = \bar{g}\bar{h}$. Finally, by Hensel's lemma, these can be lifted to a non-trivial factorization $f = gh$ for $f \in A[x]$, contradicting the irreducibility of f .

To close the argument, recall that every maximal ideal of B lies above a maximal ideal of A , but A has only one maximal ideal $\mathfrak{p}$, so B has a unique (nonzero) maximal ideal $\mathfrak{P}$. Thus B is a local Dedekind domain, hence a local PID, and not a field, which is an equivalent definition to a DVR, completing the proof.

Note that completion is necessary for this lemma to work:

Example 1.4.5: Failure Without Completion

Take $A = \mathbb{Z}_{(5)}$ which is a DVR with maximal ideal (5) but is not complete. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(i)$. Then $\mathcal{O}_L = \mathbb{Z}_{(5)}[i]$, which is a PID, but is *not* a DVR, as the ideals $(1 + 2i)$ and $(1 - 2i)$ are both maximal and not equal. However, if we take $A = \mathbb{Z}_5$ and $K = \mathbb{Q}_5$ with $L = \mathbb{Q}_5(i) = \mathbb{Q}_5$, and $B = \mathbb{Z}_5[i] = \mathbb{Z}_5$, it is a DVR, since $x^2 + 1$ has roots in $\mathbb{F}_5$ and we can lift roots in $\mathbb{Z}_5$ via Hensel's lemma.

Notice that the factorization $5 = (1 + 2i)(1 - 2i)$ in $\mathbb{Z}_{(5)}[i]$ when replaced with $\mathbb{Z}_5$ we will get that (5) is the product of the maximal ideal and a unit. In fact, no matter what prime we localize at we get this result!

This immediately gives the following

Corollary 1.4.6: The Fundamental Identity For Local Fields

Let K be a local field and L/K a finite extension. Let $\mathfrak{P}$ be the maximal prime of $\mathcal{O}_L$ and $\mathfrak{p}$ the maximal prime of K . Then:

$$[L : K] = n = ef$$

where e and f are the ramification index and inertia degree respectively.

Proof:

As there is only one prime lying above, the result follows from the global fundamental identity (see [4, chapter 22.4])

Finally, we show that completion is preserved too. First, recall that we may define a norm on a vector-space over k $\| - \| : V \rightarrow \mathbb{R}_{\geq 0}$, and that if k is complete then all norms on a finite dimensional k -vector-space are equivalent, including the maximal and euclidean norm. Now:

Theorem 1.4.7: Extension of DVR's, Completion

Let A be a complete DVR with maximal ideal $\mathfrak{p}$, $K = \text{Frac}(A)$, and $\nu_{\mathfrak{p}}$ a discrete valuation which gives the absolute value $|x|_{\mathfrak{p}} := c^{\nu_{\mathfrak{p}}(x)}$ where $0 < c < 1$. Let L/K be a finite extension of degree n . Then the following are equivalent:

1. There is a unique absolute value $|x| := |N_{L/K}(x)|_{\mathfrak{p}}^{1/n}$ on L that extends $| - |_{\mathfrak{p}}$
2. The field L is complete with respect to $| - |$, and its valuation ring $\{x \in L \mid |x| \leq 1\}$ is equal to the integral closure B of A in L

Furthermore, if L/K is separable, then B is a complete DVR whose maximal ideal $\mathfrak{P}$ induces:

$$|x| = |x|_{\mathfrak{P}} := c^{\frac{1}{e_{\mathfrak{P}}} \nu_{\mathfrak{P}}(x)}$$

where $e_{\mathfrak{P}}$ is the ramification index of $\mathfrak{P}$ (recall that $\mathfrak{p}B = \mathfrak{P}^{e_{\mathfrak{P}}}$).

Proof:

Let's first show that $|\cdot|$ is an absolute value. Certainly $|x| = 0$ if and only if $x = 0$, and $|\cdot|$ is multiplicative as the norm is. To show the triangle inequality, it suffices to show that :

$$|x| \leq 1 \Rightarrow |x+1| \leq |x| + 1$$

since in general we always have:

$$|y+z| = |z||y/z+1| \quad |y|+|z| = |z|(|y/z|+1)$$

Without loss of generality, we may assume that $|y| \leq |z|$. Then we can even show that $|x| \leq 1 \Rightarrow |x+1| \leq 1$ via;

$$\begin{aligned} |x| &\leq 1 \\ \Leftrightarrow |N_{L/K}(x)|_{\mathfrak{p}} &\leq 1 \\ \Leftrightarrow N_{L/K}(x) &\in A \\ \Leftrightarrow x &\in B \\ \Leftrightarrow x+1 &\in B \\ \Leftrightarrow |x+1| &\leq 1 \end{aligned}$$

The first bi-conditional is from the definition of $|\cdot|$, the second from the definition of $|\cdot|_{\mathfrak{p}}$, the third from lemma 1.4.3, the fourth is since it is a ring, and the fifth is from the first three by replacing x with $x+1$. This

Now, let $x \in K$. Then:

$$|x| = |N_{L/K}(x)|_{\mathfrak{p}}^{1/n} = |x^n|_{\mathfrak{p}}^{1/n} = |x|_{\mathfrak{p}}$$

showing it is indeed an extension. If $|\cdot|_{\mathfrak{p}}$ is non-trivial, then $|x|_{\mathfrak{p}} \neq 1$ for some $x \in K^*$, and if $|x|^a = |x|_{\mathfrak{p}} = |x|$, then $a = 1$, showing that $|\cdot|$ is the unique absolute value in its equivalence class extending $|\cdot|_{\mathfrak{p}}$. As every norm on L induces the same topology, $|\cdot|$ is the only absolute value on L that extends $|\cdot|_{\mathfrak{p}}$.

This covers the equivalences, so now assume L/K is separable. Then B is a DVR by theorem 1.4.4. It is complete since it is the valuation ring of L . Next, let $\mathfrak{P}$ be the unique maximal ideal of $\mathfrak{B}$. The valuation $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and so

$$\nu_{\mathfrak{P}}(x) = e_{\mathfrak{P}}\nu_{\mathfrak{p}}(x)$$

Then as $0 < c^{1/e_{\mathfrak{P}}} < 1$, we have the absolute value on L induced by $\nu_{\mathfrak{P}}$ to be:

$$|x|_{\mathfrak{P}} := (c^{1/e_{\mathfrak{P}}})^{\nu_{\mathfrak{P}}(x)}$$

This is indeed equal to $|\cdot|$, for it extends $|\cdot|_{\mathfrak{p}}$ which is known to have a unique extension. Computing:

$$y|x|_{\mathfrak{P}} = c^{\frac{1}{e_{\mathfrak{P}}}\nu_{\mathfrak{P}}(x)} = c^{\frac{1}{e_{\mathfrak{P}}}\nu_{\mathfrak{P}}(x)} = c^{\nu_{\mathfrak{p}}(x)} = |x|_{\mathfrak{p}}$$

completing the proof.

By transitivity of the norm in towers, we can uniquely extend the absolute value on the fraction field K of a complete DVR to an algebraic closure $\overline{K}$.

Corollary 1.4.8: DVR Completion And Number Rings

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Let $\mathfrak{P}/\mathfrak{p}$. Then

$$\nu_{\mathfrak{P}}(x) = \frac{1}{f_{\mathfrak{P}}} \nu_{\mathfrak{p}}(N_{L/K}(x))$$

for all $x \in L$

Proof:

$$\nu_{\mathfrak{p}}(N_{L/K}(x)) = \nu_{\mathfrak{p}}(N_{L/K}(\bar{x})) = \nu_{\mathfrak{p}}(N_{L/K}(\mathfrak{P}^{\nu_{\mathfrak{P}}(x)})) = \nu_{\mathfrak{p}}(\mathfrak{p}^{f_{\mathfrak{P}} \nu_{\mathfrak{P}}(x)}) = f_{\mathfrak{P}} \nu_{\mathfrak{P}}(x)$$

1.5 Dedekind-Kummer for local rings

Recall the Dedekind-Kummer Theorem ([4, chapter 22.4]) where a prime $\mathfrak{P}/\mathfrak{p}$ factors like the minimal polynomial of the finite separable extension L/K (i.e. $AKLB$) if the ring of integers of L is monogenic or the prime doesn't contain the conductor. In the case where the ring of integers of K is a DVR and the residue of the extension is finite, we get that B is always monogenic. This requires Nakayama's lemma which we remind the reader of:

Lemma 1.5.1: Nakayama's Lemma

Let A be a local ring with maximal ideal $\mathfrak{p}$ and let M be a finitely generated A -module. If the image of $x_1, \dots, x_n \in M$ generate $M/\mathfrak{p}M$ as an $(A/\mathfrak{p})$ -vector space, then $x_1, \dots, x_n$ generate M as an A -module.

Proof:

see [4, section 20.5.5]

This gives a local version of the Dedekind-Kummer theorem that doesn't even need that A, B be Dedekind domain

Corollary 1.5.2: Local Ring Maximal Ideal Extension

Let A be a local Noetherian ring with maximal ideal $\mathfrak{p}$, $g \in A[x]$ be a polynomial with quotient $\bar{g} \in (A/\mathfrak{p})[x]$, and let α be the image of x in the ring $B := A[x]/(g(x))$. Then the maximal ideals of B are $(\mathfrak{p}, g_i(\alpha))$ where $g_1, \dots, g_m \in A[x]$ are lifts of the distinct irreducible polynomials $\bar{g}_i \in (A/\mathfrak{p})[x]$ that divide $\bar{g}$.

Nakayama will give the correspondence, and the fact that A has only a single maximal ideal will force that the extension have only so many possible maximal ideals:

Proof:

By Nakayama's lemma, the quotient map $B \rightarrow B/\mathfrak{p}B$ gives a one-to-one correspondence between the maximal ideals of B and the maximal ideals of $B/\mathfrak{p}B$, and hence:

$$B/\mathfrak{p}B \cong A[x]/(\mathfrak{p}, g(x)) \cong (A/\mathfrak{p})[x]/(\bar{g}(x))$$

Now, each maximal ideal of the right hand side is the quotient of an irreducible divisor of $\bar{g}$ (you can use the Chinese remainder theorem if you're unconvinced) and hence as $(A/\mathfrak{p})[x]$ is a PID it is one of the $\bar{g}_i$, as we sought to show.

Theorem 1.5.3: Local Dedekind-Kummer Theorem

Take $AKLB$, where A, B are DVR's, and let $\kappa = A/\mathfrak{p}, \lambda = B/\mathfrak{P}$ be the residue fields given a prime $\mathfrak{P}$ over $\mathfrak{p}$. Then if λ/κ is separable,

$$B = A[\alpha]$$

for appropriate $\alpha \in B$. If L/K is unramified, this then holds for every lift α for any generator $\bar{\alpha}$ for $\lambda = \kappa(\bar{\alpha})$.

Hence, by [4, Corollary 22.4.8], every ring of integers extension $\mathcal{O}_L/\mathcal{O}_K$ is monogenic, and the [global] Dedekind-Kummer Theorem applies.

Proof:

Let $\mathfrak{p}B = \mathfrak{P}^e$ be the factorization of $\mathfrak{P}/\mathfrak{p}$ and let $f = [\lambda : \kappa]$ be the inertia degree so that $ef = n := [L : K]$. As the extension λ/κ is separable, there exists a primitive element

$$\lambda = \kappa(\bar{\alpha}_0)$$

for some $\bar{\alpha}_0 \in \lambda$ whose minimal polynomial $\bar{g}$ is separable of degree f . Let $g \in A[x]$ be the monic lift of $\bar{g}$ and let α_0 be any lift of $\bar{\alpha}_0$ to B . If $\nu_{\mathfrak{P}}(g(\alpha_0)) = 1$, let $\alpha := \alpha_0$. Otherwise, let $\pi \neq 0$ be any uniformizer for B and let $\alpha := \alpha_0 + \pi_0 \in B$. Then $\alpha \equiv \alpha_0 \pmod{\mathfrak{P}}$. Decomposing

$$g(x + \pi_0) = g(x) + \pi_0 g'(x) + \pi_0^2 h(x)$$

for appropriate $h \in A[x]$, we get:

$$\nu_{\mathfrak{P}}(g(\alpha)) = \nu_{\mathfrak{P}}(g(\alpha_0 + \pi_0)) = \nu_F B(g(x + \pi_0) = g(x) + \pi_0 g'(x) + \pi_0^2 h(x)) = 1$$

and hence $\pi := g(\alpha)$ is the uniformizer of B .

Now, we claim that $B = A[\alpha]$. This is the same as $1, \alpha, \dots, \alpha^{n-1}$ generates B as an A -module. By Nakayama's lemma, this is equivalent to asking if $1, \alpha, \dots, \alpha^{n-1}$ span $B/\mathfrak{p}B$ as a κ -vector space. As $\mathfrak{p} = \mathfrak{P}^e$, $\mathfrak{p}B = (\pi^e)$. This means we can represent each element of $B/\mathfrak{p}B$ as cosets:

$$b + \mathfrak{p}B = b_0 + b_1\pi + b_2\pi^2 + \dots + b_{e-1}\pi^{e-1} + \mathfrak{p}B$$

where $b_0, \dots, b_{e-1}$ are given up to equivalence modulo πB . Now the field theory $1, \bar{\alpha}, \dots, \bar{\alpha}^{f-1}$

forms a basis for $B/\pi B = B/\mathfrak{P}$ as a κ -vector space, so we can write this as:

$$\begin{aligned} b + \mathfrak{p}B &= (a_0 + a_1\alpha + \cdots + a_{f-1}\alpha^{f-1}) \\ &\quad + (a_f + a_{f+1}\alpha + \cdots + a_{2f-1}\alpha^{f-1})g(\alpha) \\ &\quad + \cdots \\ &\quad + (a_{ef-f+1} + a_{ef-f+2}\alpha + \cdots + a_{ef-1}\alpha^{f-1})g(\alpha)^{e-1} + \mathfrak{p}B \end{aligned}$$

As $\deg g = f$ and $n = ef$, this expresses $b + \mathfrak{p}B$ in the form $b' + \mathfrak{p}B$ with b' in the A -span of $1, \dots, \alpha^{n-1}$. Thus:

$$B = A[\alpha]$$

Finally, if L/K is unramified, then λ/κ is separable, and $e = 1$, $f = n$, and so there is no need to check if $g(\alpha)$ is a uniformizer which immediately gives $\alpha = \alpha_0$, as we sought to show.

1.6 Ramification in Local Fields

Let L/K be a finite extension of local fields. This section shall explore how to decompose such an extension into an unramified extension and a completely ramified extension. To remind the reader:

Definition 1.6.1: Ramification Index And Inertia Degree Of Local Fields

Let K be a local field and L/K a finite extension. Let ν be the discrete valuation associated to K and w be extension of ν given by theorem 1.4.2. Then as $\nu(K^*) \subseteq w(L^*)$ define the *ramification index* of L with respect to K to be:

$$e(L : K) = [w(L^*) : \nu(K^*)]$$

and the *inertia degree* of L with respect to K to be

$$f(L : K) = [\lambda : \kappa]$$

where λ, κ are the residue fields of L and K respectively.

Notice that $w = \frac{1}{n}\nu \circ N_{L/K}$, and hence is discrete. If $\mathfrak{o}, \mathfrak{p}, \pi$ and $\mathcal{O}, \mathfrak{P}, \Pi$ are the valuation-ring/maximal-ideal/prime-element of K and L respectively, then:

$$e(L : K) = [w(\Pi)\mathbb{Z} : \nu(\pi)\mathbb{Z}]$$

Hence $\nu(\pi) = ew(\Pi)$ which gives:

$$\pi = u\Pi^e$$

for some unit $u \in \mathcal{O}^*$, which mirrors the result for the definition of ramification index in the case of global fields.

1.6.1 Unramified Extensions

With the fundamental identity property re-defined, let us upgrade the definition of unramified extensions from [4, chapter 22.4]:

Definition 1.6.2: Unramified Extension for local field

Let A be a complete DVR with finite residue field, K its field of fractions, and L a finite extension. Then if:

$$[L : K] = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})] = f$$

Then L/K is said to be an *unramified extension*.

Hence, when L/K is finite and unramified, the degree n of L/K has the same degree as the finite separable extension λ/κ

$$[L : K] = [\lambda : \kappa]$$

We may thus ask what is the relation of L/K and λ/κ . In particular, if we fix K with residue field κ , what is the relationship between the finite unramified extensions L/K of degree n and the finite separable extensions λ/κ of degree n . We already know that each L/K uniquely determined a corresponding λ/κ , but is the converse true too? In fact, it is almost as nice as it gets:

Theorem 1.6.3: Unramified Extension And Residue Field Category Equivalence

Let A be a complete DVR with $K = \text{Frac}(A)$ its residue field. Then the category $\mathcal{C}_K^{nr}$ and $\mathcal{C}_\kappa^{\text{sep}}$ are equivalent; the functor $\mathcal{F} : \mathcal{C}_K^{nr} \rightarrow \mathcal{C}_\kappa^{\text{sep}}$ sends each unramified extension L/K to its residue field λ , and each K -algebra homomorphism $\varphi : L_1 \rightarrow L_2$ to the κ -algebra homomorphism $\bar{\varphi} : \lambda_1 \rightarrow \lambda_2$ given by $\bar{\varphi}(\bar{\alpha}) := \overline{\varphi(\alpha)}$ where α is any lift of $\bar{\alpha} \in \lambda_1 = B_1/\mathfrak{P}_1$ to B_1 and $\bar{\varphi}$ is the reduction of $\varphi(\alpha) \in B_2$ to $\lambda_2 := B_2/\mathfrak{P}_2$.

This means that $\mathcal{F}$ gives a bijection between the isomorphism classes in $\mathcal{C}_K^{nr}$, and $\mathcal{C}_\kappa^{\text{sep}}$ and if L_1, L_2 have residue fields λ_1, λ_2 , then $\mathcal{F}$ induces a bijection of finite sets:

$$\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_\kappa(\lambda_1, \lambda_2)$$

The most interesting part of the proof is $\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_\kappa(\lambda_1, \lambda_2)$, which will ultimately come down to Hensel's lemma after all the appropriate build-up.

Proof:

Let us start with showing that $\mathcal{F}$ is well-defined. The object certainly map to the object, so we need to show that the morphism map is independent of lifts. Let $\varphi : L_1 \rightarrow L_2$ be a K -algebra homomorphism, and for $\bar{\alpha} \in \lambda_1$, let α, α' be two lifts to B_1 . Then $\alpha - \alpha' \in \mathfrak{P}_1$, implying $\varphi(\alpha - \alpha') \in \varphi(\mathfrak{P}_1) \stackrel{!}{=} \varphi(B_1) \cap \mathfrak{P}_2 \subseteq \mathfrak{P}_2$, and hence $\overline{\varphi(\alpha)} = \overline{\varphi(\alpha')}$, showing it is well-defined. The $\stackrel{!}{=}$ equality comes from φ restricts to the ring homomorphism $B_1 \rightarrow B_2$ and $B_2/\varphi(B_1)$ is a finite extension of DVRs in which $\mathfrak{P}_2$ lies over the prime $\varphi(\mathfrak{P}_1)$ of $\varphi(B_1)$. Checking that the identity is set to the identity and composition is well-defined now follows quickly.

Let us now show that $\mathcal{F}$ gives an equivalence of categories. Recall from [4, Part 8] that this means that:

1. $\mathcal{F}$ is essentially surjective: each separable extension λ/κ is isomorphic to the residue field of some unramified L/K
2. $\mathcal{F}$ is full and faithful, i.e. the hom-map is a bijection.

Starting with essential surjectivity, given a finite separable extension λ/κ , we may apply the primitive element theorem and get:

$$\lambda \cong \kappa(\bar{\alpha}) \cong \frac{\kappa[x]}{(\bar{g}(x))}$$

For appropriate $\bar{\alpha} \in \lambda$ whose minimal polynomial $\bar{g}$ is necessarily monic, irreducible, separable, and of degree $n := [\lambda : \kappa]$. Take any monic lift $g \in A[x]$. Then g is also irreducible, separable, and of degree n . Now let :

$$L := \frac{K[x]}{(g(x))} = K(\alpha)$$

Then L/K is a finite separable extensions. By corollary 1.5.2 $(\mathfrak{p}, g(\alpha))$ is the unique maximal ideal of $A[\alpha]$, and :

$$B/\mathfrak{P} \cong \frac{A[\alpha]}{(\mathfrak{p}, g(\alpha))} \cong \frac{A[x]}{(\mathfrak{p}, g(x))} \cong \frac{(A/\mathfrak{p})[x]}{(\bar{g}(x))} \cong \lambda$$

and hence:

$$[L : K] = \deg g = [\lambda : \kappa] = n = [f]$$

showing that L/K is unramified, namely by the fundamental identify the ramification index of $\mathfrak{P}$ is necessarily 1.

Next, to show that $\mathcal{F}$ is full and faithful, let us start with the induced maps:

$$\mathrm{Hom}_K(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_A(B_1, B_2) \rightarrow \mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$$

The first map was shown in [4, Chapter 22.1]. It must be shown that the same applies for the second map.

First, notice that:

$$\mathrm{Hom}_A(B_1, B_2) = \mathrm{Hom}_A\left(\frac{A[x]}{(g(x))}, B_2\right) = \mathrm{Hom}_A(A(\alpha), B_2)$$

which comes from B_1 always being a monogenic extension of A (Theorem 1.5.3) and that each A -module homomorphism is uniquely determined by the image of x in B_2 . This gives a bijection between $\mathrm{Hom}_A(B_1, B_2)$ and the roots of g in B_2 . Now, with the right choice of α , we can give the map. By the primitive root theorem, $\lambda_1 = \kappa(\bar{\alpha})$. Lift $\bar{\alpha}$ to $\alpha \in B_1$. Then $L_1 = K(\alpha)$ since $[L_1 : K] = [\lambda_1 : \kappa]$, and these two quantities equal to the degree of the minimal polynomial of $\bar{g}$ of $\bar{\alpha}$ which cannot be less than the degree of the minimal polynomial g of α (both are monic, so no issues will come from characteristics). Then we saw in theorem 1.5.3 that $B_1 = A[\alpha]$. Now, consider:

$$\mathrm{Hom}_\kappa(\lambda_1, \lambda_2) = \mathrm{Hom}_\kappa\left(\frac{\kappa[x]}{(\bar{g}(x))}, \lambda_2\right) = \mathrm{Hom}_\kappa(\kappa(\bar{\alpha}), \lambda_2)$$

where the equality again comes from the uniquely determined image of x in λ_2 , and there is a bijection between $\mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$ and the roots of $\bar{g}$ in λ_2 . Now, $\bar{g}$ is separable, so by Hensel's lemma every root of $\bar{g}$ in $\lambda_2 = B_2/\mathfrak{P}_2$ lifts to unique roots of g in B_2 . But then the map is a bijection, as we sought to show.

Note that for the bijection it was only necessary that L_1/K be unramified, so at least the bijection holds if L_2/K is unramified. Furthermore, using this bijection we can compute the galois group of

automorphisms of a field by looking at the automorphisms of the residue field.

Corollary 1.6.4: Characterizing Unramified Extensions through residue field

Assume $AKLB$ where A is a complete DVR with residue field κ . Then L/K is unramified if and only if $B = A[\alpha]$ for some $\alpha \in L$ whose minimal polynomial has separable image in $\kappa[x]$.

Proof:

The “if” was given by theorem 1.6.3. For the reverse, note that if g is the minimal polynomial, $\bar{g}$ must be irreducible, otherwise Hensel’s lemma would give a lift to a non-trivial factorization of $\bar{g}$, so the residue field extension is separable and has the same degree as L/K , and hence L/K is unramified as $[L : K] = f$, completing the proof.

We shall now build-up to show that all unramified extensions where the residue field κ is finite will be isomorphic to a Cyclotomic extension. Starting with the following:

Lemma 1.6.5: Unramified Cyclotomic Extensions

Let A be a complete DVR with fraction field K and residue field κ , and let ζ_n be the primitive n th roots of unity in some algebraic closure of K with $(n, p) = 1$ where $\text{char } \kappa = p$. Then $K(\zeta_n)/K$ is unramified

Proof:

By the relation’s of the root of $f(x) = x^n - 1$, The field $K(\zeta_n)$ is splitting field of this polynomial over K , The image $\bar{f}$ is separable when $\nmid n$ as $\gcd(\bar{f}, \bar{f}') \neq 1$ if and only if $\bar{f}' = nx^{n-1}$ is zero, or equivalently when $p \mid n$. When $\bar{f}$ is separable, so are all of its divisors, including the reduction of the minimal polynomial of ζ_n (which is a factor of f after dividing by $x - 1$), and it must be irreducible since otherwise we would contain a non-trivial factorization by Hensel’s lemma. Thus, the residue field $\kappa(\zeta_n)$ is a separable extension of κ , and so by corollary 1.6.4 $K(\zeta_n)/K$ is unramified, as we sought to show.

Adding on the extra condition that the residue field is finite (which is always the case for local fields), we get to characterize unramified extensions:

Proposition 1.6.6: Characterizing Unramified Extension Of Local Fields

Let A be a complete DVR with fraction field K and finite residue field $\mathbb{F}_q$. Let L/K be a degree n extension. Then L/K is unramified iff

$$L \cong K(\zeta_{q^n-1})$$

Furthermore, $A[\zeta_{q^n-1}]$ is the integral closure of A in L , and L/K is a Galois extension with

$$\text{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

Proof:

The above corollary showed the “only if” direction, so let us show the “if direction” by assuming L/K is unramified. Then $[\lambda : \kappa] = [L : K] = n$ where by the finiteness of the residue field $\lambda = \mathbb{F}_{q^n}$. By field theory, $\mathbb{F}_{q^n}$ has the multiplicative group of order $q^n - 1$ generated by some $\bar{\alpha}$, say $\bar{g}$ is its minimal polynomial. Then $\bar{g}$ divides $x^{q^n-1} - 1$, and since $\bar{g}$ is irreducible, $(\bar{g})$ is coprime to $(x^{q^n-1} - 1)/\bar{g}$. By Hensel’s lemma, we can lift this all to $A[\zeta_{q^n-1}]$ and show that g divides $x^{q^n-1} - 1 \in A[x]$, and again by Hensel’s lemma we can lift $\bar{\alpha}$ to a root α of g which shows that α is a root of $x^{q^n-1} - 1$. Thus, it is a primitive $(q^n - 1)$ -root of unity as the reduction is, which implies that

$$L \cong K(\zeta_{q^n-1})$$

Finally, by theorem 1.5.3 we have $B \cong A[\zeta_{q^n} - 1]$ and L is the splitting field of $x^{q^n-1} - 1$ (we can lift the factorization of this polynomial from $\mathbb{F}_{q^n}$ to L via Hensel’s lemma). Hence L/K is galois, and the bijection between the $(q^n - 1)$ -roots of unity in L and $\mathbb{F}_{q^n}$ induces $\text{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$, as we sought to show.

The fact that an extension is unramified if and only if $L \cong K(\zeta_{q^n-1})$ implies that we can also deduce when a cyclic extension *is* ramified:

Corollary 1.6.7: Ramification Of Cyclic Extension

Let A be a complete DVR with field of fraction K and finite residue field of characteristic p . Suppose that K does not contain a primitive p th root of unity. Then the extension $K(\zeta_m)/K$ is ramified if and only if $p|m$.

Proof:

If $p \nmid m$, by lemma 1.6.5 $K(\zeta_m)/K$ is unramified. if $p|m$, then $K(\zeta_m)$ must contain $K(\zeta_p)$ which by proposition 1.6.6 is unramified if and only if $K(\zeta_p) \cong K(\zeta_{p^n-1})$ with $n = [K(\zeta_p) : K]$, which occurs if $p|p^n - 1$ (since $\zeta_p \notin K$ by assumption), which certainly is not the case. Hence, $K(\zeta_p)$ is ramified, implying $K(\zeta_m)$ is ramified, as we sought to show.

Now with a choice of complete DVR, we can fully understand the unramified extensions

Example 1.6.8: Local Number Fields

Let $A = \mathbb{Z}_p$ so that $K = \mathbb{Q}_p$ and $\kappa = \mathbb{F}_p$. Then for each n , the galois field $\mathbb{F}_{p^n}$ is the unique extension of degree n . Thus, by the above results, there is a unique unramified extension of degree n , $L/\mathbb{Q}_p$. This can be explicitly constructed by adjoining a primitive root of unity ζ_{p^n-1} to $\mathbb{Q}_p$.

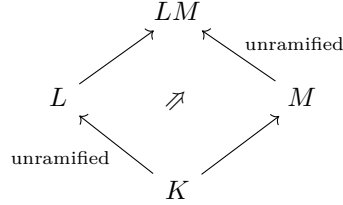
Maximal Unramified Extension

Let us build-up to the notion of maximal unramified extension, which is an important object for the Artin map. First, the “3rd isomorphism theorem” property applies for unramified extensions

Proposition 1.6.9: Unramified Square

Let L/K and M/K be two extensions, with $\mathfrak{p} \subseteq K$, $\mathfrak{P} \subseteq L$, $\mathfrak{P}' \subseteq M$, $\mathfrak{P} \cap K = \mathfrak{P}' \cap K = \mathfrak{p}$. Then if L/K is unramified for $\mathfrak{P}$ over $\mathfrak{p}$ if LM/M is unramified for $\mathfrak{P}'$ over $\mathfrak{p}$.

Visually,

**Proof:**

Define:

$$\mathcal{O}, \mathfrak{p}, \kappa' \quad \mathcal{O}, \mathfrak{P}, \lambda, \quad \mathcal{O}', \mathfrak{p}', \kappa \quad \mathcal{O}', \mathfrak{P}', \lambda'$$

to the ring of integers, prime, and residue field of K, L, M, LM respectively, and assume L/K is finite. Then λ/κ is also finite and separable, and hence generated by a primitive element $\bar{\alpha}$, $\lambda = \kappa(\bar{\alpha})$. Without loss of generality, we may have $\alpha \in \mathcal{O}$ by multiplying out the denominator, and hence there is a minimal polynomial for α , $f(x) \in \mathcal{O}[x]$. Let $\bar{f} \in \kappa[x]$ be $\bar{f}(x) = f(x) \bmod \mathfrak{p}$. By minimality, the minimal polynomial of α in λ divides $\bar{f}$, hence:

$$[\lambda : \kappa] \leq \deg(\bar{f}) = \deg(f) = [K(\alpha) : K] \leq [L : K] = [\lambda : \kappa]$$

hence $[\lambda : \kappa] \leq [K(\alpha) : K] \leq [\lambda : \kappa]$, and so $L = K(\alpha)$, which also shows that $\bar{f}$ is the minimal polynomial of $\bar{\alpha}$ over κ .

Thus, $LM = M(\alpha)$. To show this extension is unramified, let $g(x) \in \mathcal{O}'[x]$ be the minimal polynomial of α over κ' , and let $\bar{g}(x) \equiv g(x) \bmod \mathfrak{p}' \in \kappa'[x]$. Then as a factor of $\bar{f}$, it is separable, hence irreducible over κ' (or else $g(x)$ is reducible by Hensel's lemma!). Thus we get:

$$[\lambda' : \kappa'] \leq [LM : M] = \deg(g) = \deg(\bar{g}) = [\kappa'(\bar{\alpha}) : \kappa'] \leq [\lambda' : \kappa']$$

Giving that LM/M is unramified.

Finally, if L/K is a extension of the unramified extension LM/K , then it follows that LM/L is unramified, and thus so is L/K by the degree formula, as we sought to show.

Corollary 1.6.10: Composite Of Unramified Extensions

The composite of two unramified extensions of L is again unramified

Proof:

It suffices to show that if L/K , M/K , namely as L/K is unramified by proposition 1.6.9 LM/M is unramified, and hence LM/K is unramified as separability is transitive and the degrees are multiplicative.

Hence, the following is well-defined:

Definition 1.6.11: Maximally Unramified Extension (Hilbert Class Field)

Let L/K be an algebraic extensions. Then the composite of all unramified extensions M/K is called the *maximally unramified extension* of L/K .

If $\bar{K}$ is the algebraic closure of K , then K_{nr}/K is the maximal unramified extension of K .

The term K_{nr} is given by the french “non-ramifiée”.

Example 1.6.12: Maximal Unramified Extension For Local Fields

Take $K = \mathbb{Q}_p$. Then $\mathbb{Q}_p^{nr}$ will be an finite extension with galois group:

$$\text{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p^{nr}) \cong \text{Gal}_{\mathbb{F}_p}(\bar{\mathbb{F}}_f) = \varprojlim_n \text{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n}) \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} =: \hat{\mathbb{Z}}$$

where the last ring is called the *profinite completion* of $\mathbb{Z}$. The field has value group $\mathbb{Z}$ (it's image under valuation) and residue field $\bar{\mathbb{F}}_p$.

Corollary 1.6.13: Root Of Unity In Maximally Unramified Extension

Let K_{nr}/K be the maximally unramified extension. Then K_{nr} contains all the roots of unity of order M not divisible by the characteristic κ

If κ is a finite field, then the extension K_{nr}/K is generated by the roots of unity.

Proof:

This is putting together the results we saw earlier. The polynomial $x^m - 1$ splits over $\bar{\kappa}_s$, and hence over K_{nr} by Hensel's lemma. Similarly, as the roots of unity generate $\bar{\kappa}_s/\kappa$, by Hensel's they also generate K_{nr}

This also shows then that K_{nr} is a Cyclotomic extension of K .

Proposition 1.6.14: Residue Field Of Maximally Unramified Extension

Let M/K be the maximal unramified extension of K . Then the residue class field of M is the separable closure of λ_s of κ .

Proof:

This follows from the equivalence of categories presented earlier.

To wrap up this section, let us now look at decomposing an extension L/K into the ramified and unramified portion, and in the case where L/K is galois to find what it is:

Assume $AKLB$ with A a complete DVR and separable residue field extension λ/κ . Let e, f be the ramification index and residue field degree, and let $\mathfrak{P}$ be the unique prime of B . Then:

1. There is a unique intermediate field extension E/K that contains every unramified extension of K in L and has degree $[E : K] = f$
2. The extension L/E is totally ramified and has degree $[L : E] = e$
3. If L/K is galois, then $\text{Gal}_K(L)$ is the decomposition group of $D_{\mathfrak{P}}$. $\text{Gal}_E(L)$ is the inertia subgroup of $I_{\mathfrak{P}}$, and E/K is galois with $\text{Gal}_K(E) \cong D_{\mathfrak{P}}/I_{\mathfrak{P}} \cong \text{Gal}_{\kappa}(\lambda)$

1. exercise
2. tower argument
3. Take $D_{\mathfrak{P}} \subseteq \text{Gal}_K(L)$. Then as there is only one prime, the entire group must be the stabilizer, and so $D_{\mathfrak{P}} = \text{Gal}_K(L)$. Take now $\mathfrak{P}_E = \mathfrak{P} \cap E$. Then recall that we get:

then these three groups all have order e , and will then have to be equal by their proprieties. As $I_{\mathfrak{P}}$ is a normal subgroup of $D_{\mathfrak{P}}$, E/K is normal, and as it is separable it is Galois. It follows then that:

completing the proof. $\square$

Global Case:

$$\left\{ \begin{array}{l} K = K^{\mathbb{F}_3} \\ e \mid K \\ f \mid K \\ g \mid K \\ Q = K^{\text{Gal}(K/\mathbb{Q})} \end{array} \right. \begin{array}{l} I(y/p) \\ I(y/p) \\ I(y/p) \\ I(y/p) \end{array}$$

Local case:

$$\left\{ \begin{array}{l} K_p \\ e \mid K_p \\ f \mid K_p \\ Q_p = K_p^{\text{Gal}(K_p/\mathbb{Q}_p)} \end{array} \right. \begin{array}{l} I(y/p) \\ I(y/p) \\ I(y/p) \\ \text{Gal}(K_p/\mathbb{Q}_p) \end{array}$$

Notation: If $e=1$, K_p is unramified over $\mathbb{Q}_p$.
If $f=1$, K_p is totally ramified over $\mathbb{Q}_p$.

Nathanael Chwojko-Srawley

Exercise 1.6.1

1. Assume $AKLB$, with A a complete DVR with finite residue field. Then L/K is unramified if and only if $N_{L/K}(B^*) = A^*$. Is this the case for global fields?

1.6.2 Ramified Extensions

We are now in a position where if we have any finite separable extension L/K where A is a complete DVR, then we have a tower $L/E/K$ where E/K is unramified and L/E is totally ramified. We just classified the unramified extensions (or in particular the unramified extensions of field of fractions of complete DVRs), so let us now focus on the ramified case.

Totally ramified extensions of complete DVR

Recall that a polynomial is *Eisenstein* if for some prime $\mathfrak{p} \subseteq R$, the polynomial $f(x) \in R[x]$ has coefficients $a_i \in \mathfrak{p}$ but $a_i^2 \notin \mathfrak{p}$. In the case where the ring is a DVR, this implies that a_0 is a uniformizer, as $\nu_{\mathfrak{p}}(a_0) = 1$. As the ring is a DVR, it is a PID and hence a UFD, meaning we can apply Gauss's lemma and get that it is irreducible over the field of fractions.

Lemma 1.6.16: DVR Extension Using Eisenstein Polynomial

Let A be a DVR and let $f \in A[x]$ be an Eisenstein polynomial. Then $B = A[\pi] = A[x]/(f)$ is a DVR with uniformizer π , where π is the image of x in $A[x]/(f)$.

Proof:

Let $\mathfrak{p}$ be the maximal ideal of A . Then $f \equiv x^n \pmod{\mathfrak{p}}$, and so by corollary 1.5.2 the ideal $\mathfrak{P}(\mathfrak{p}, x) = (\mathfrak{p}\pi)$ is the only maximal ideal of B . Let $f = \sum_i a_i x^i$. Then as f is Eisenstein, $\mathfrak{p} = (a_0)$, $\mathfrak{P} = (a_0, \pi)$ and $a_0 = -a_1\pi - f_2\pi^2 - \dots - \pi^n \in (\pi)$, hence $\mathfrak{P} = (\pi)$. As the unique maximal ideal is principal, B is a DVR as we sought to show.

Theorem 1.6.17: Characterizing Totally Ramified Extensions

Assume $AKLB$ with A a complete DVR and π a uniformizer for B . Then the extension L/K is totally ramified if and only if $B = A[\pi]$ and the minimal polynomial of π is Eisenstein

Proof:

By proposition theorem 1.5.3, $B = A[\pi]$. Let f be the minimal polynomial of π . If f is Eisenstein then by lemma 1.6.16 we have $\mathfrak{P} = \mathfrak{p}^n$, hence $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}} = n$ and L/K is totally ramified.

Conversely, if L/K is totally ramified, then $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index n , implying

$$\nu_{\mathfrak{P}}(K) = n\mathbb{Z}$$

Let's first find a K -basis for L . Take the set $\{1, \pi, \pi^2, \dots, \pi^{n-1}\}$ where $\pi \in K$ is a uniformizer given by $\nu_{\mathfrak{p}}$. This set is linearly independent as their valuations are distinct modulo $\nu_{\mathfrak{P}}(K)$, namely

if $\sum_{i=0}^{n-1} a_0 \pi^i = 0$ then it must be the case that for some nonzero a_i, a_j $\nu_{\mathfrak{P}}(a_i \pi^i) = \nu_F B(a_j \pi^j)$ which is impossible. As this also spans L , we have that

$$L = K(\pi)$$

Let $f = \sum_{i=0}^n a - ix^i \in A[x]$ be the minimal polynomial of π . We must show it is Eisenstein. First note that $\nu_{\mathfrak{P}}(f(\pi)) = 0$ and $\nu_{\mathfrak{P}}(a_i \pi^i) = i \bmod n$ for $0 \leq i \leq n$. This is possible only if

$$\nu_{\mathfrak{P}}(a_0) = \nu_{\mathfrak{P}}(a_i \pi^0) = \nu_{\mathfrak{P}}(a_n \pi^n) = \nu_{\mathfrak{P}}(\pi^n) = n$$

and $\nu_{\mathfrak{P}}(a_i) \geq n$ for $0 \leq i < n$. But then this implies $\nu_{\mathfrak{p}}(a_0) = 1$ as $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index n , and $\nu_{\mathfrak{p}}(a_i) \geq 1$ for $0 \leq i < n$. But then f is Eisenstein. Then lemma 1.6.16 implies $A[\pi] \subseteq B$ is a DVR, and by maximality we get $B = A[\pi]$, completing the proof.

Example 1.6.18: Environment Title

I'd like to add example 11.6. I want to put in the exercise fomr the HW somewhere because it is actually a really cool exercise.

Definition 1.6.19: Tamely Ramified And Wildly Ramified

Assume $AKLB$ with A a complete DVR and a separable field extension of characteristic $p \geq 0$. Then:

1. The extension L/K is *tamely ramified* if $p \nmid e_{L/K}$ (which is always true if $p = 0$)
2. The extension L/K is *wildly ramified* if $p | e_{L/K}$

All unramified extensions are tamely ramified. A totally ramified extension L/K is totally tamely ramified if $p \nmid e_{L/K}$ and is *totally wildly ramified* if $e_{L/K}$ is a power of p . Note that a totally ramified extension that is wildly ramified need not be totally wildly ramified.

Proposition 1.6.20: Tower of Ramified Extensions

The tower of extensions of fields of fractions of complete DVR's with separable residue field extensions of all types of ramified extensions are transitive

Proof:

exercise

The same is not the case for compositum's, for example the compositum of totally ramified extensions need not be totally ramified (see example 1.6.18).

Theorem 1.6.21: Characterizing Totally Tamely Ramified Extensions

Assume $AKLB$ with A a complete DVR and separable residue field extension of characteristic $p \geq 0$ not dividing $n := [L : K]$. Then the extension L/K is *totally tamely ramified* if and only if $L = K(\pi_A^{1/n})$ for some uniformizer π_A of A .

The main place the tame assumption shall come is when we need to use Hensel's lemma to lift a certain solution. The equivalent of this result in the global case required that the extension be a *Kummer Extension*, which is an extension with enough roots of unity and that there are certain properties satisfied by the galois group of K .

Proof:

If $L = K(\pi_A^{1/n})$, then $\pi = \pi_A^{1/n}$ has minimal polynomial $x^n - \pi_A$, which is Eisenstein and so $A[\pi]$ is a DVR by lemma 1.6.16. By maximality of DVR's, $B = A[\pi]$, and by theorem 1.6.17 L/K is totally tamely ramified, as $p \nmid n$.

Conversely, assume L/K is totally tamely ramified and $\mathfrak{p}, \mathfrak{P}$ are the maximal ideals of A, B with uniformizer π_A, π_B respectively. Then $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}} = n$ and $\nu_{\mathfrak{P}}(\pi_B^n) = n = \nu_{\mathfrak{P}}(\pi_A)$. Then $\pi_B^n = u\pi_A$ for some unit $u \in B^*$. As we are totally ramified, $f_{\mathfrak{P}} = 1$, so B and A have the same residue field, and so if we lift the image of $u \in B/\mathfrak{P} \cong A/\mathfrak{p}$, to a unit $u_A \in A$, and replace π_A with $u_A^{-1}\pi_A$, we can assume $u \equiv 1 \pmod{\mathfrak{P}}$ (i.e. it is a unit also in $B/\mathfrak{P}$). From this, define

$$g(x) = x^n - u \in B[x] \quad \Rightarrow \quad \bar{g}(x) = x^n - 1 \in (B/\mathfrak{P})[x]$$

Then $\bar{g}'(1) = n \neq 0$ as $p \nmid n$, showing it is a primitive polynomial and so by Hensel's lemma we can lift the root of 1 to r for $g(x)$ in B . Say $\pi = \pi_B/r$. Then π is a uniformizer for B , hence by theorem 1.6.17 $B = A[\pi]$, so $L = K(\pi)$. As $\pi^n = \pi_B^n/r = \pi_B^n/u = \pi_A$, we have:

$$L = K(\pi_A^{1/n})$$

we completed the other direction, as we sought to show.

Proposition 1.6.22: Tower of Ramification

Let K be the field of fractions of a complete DVR and L/K a totally ramified extension. Then there is a unique intermediate field E/K that is totally tamely ramified and L/E is totally wildly ramified

Proof:

See MIT notes p.101

Corollary 1.6.23: Tower of Ramification in Separable Case

Let K be the field of fractions of a complete DVR with separable residue field and L/K be a finite separable extension with separable residue field extensions. Then there is a unique intermediate field E such that E/K is tamely ramified and L/E is totally wildly ramified.

Proof:

See MIT notes p.101

Exercise 1.6.2

1. Let L/K be a finite extension. Then L/K is a tamely ramified extension (not necessarily totally tamely ramified) if and only if $L = K(\sqrt[m_1]{a_1}, \dots, \sqrt[m_r]{a_r})$ where $\gcd(m_i, p) = 1$ (generalize theorem 1.6.21).
2. Generalize proposition 1.6.22 to tamely ramified extensions. Use this to show there exists a maximally tamely ramified extension.

1.7 Analysis interlude: Krasner's Lemma

Recall that if $f \in \mathbb{R}[x]$ or $\mathbb{C}[x]$ is a separable polynomial, then it is always possible to “jiggle” the roots around and f remains separable, or similarly the coefficients of f can be “jiggled” and the polynomial remains separable. In fact, more is true: the coefficients can be jiggled and the field extension $K(f(\alpha))/K$ and $K(g(\alpha))/K$ (for appropriate $K = \text{Frac}(R)$ for complete DVR R) are isomorphic. This is the result aimed to be proven in this section. Such results shall come up a few times in the book it is required to use a separable polynomial f which has certain necessary roots but has some “bad” property by making a bad choice (whether a forced choice or arbitrary choice), and so another polynomial with the desired properties is found.

Lemma 1.7.1: Automorphism Preserves Absolute Value

Let K be a fraction field of a complete DVR with algebraic closure $\bar{K}$, and extend the absolute value $|\cdot|$ to $\bar{K}$. Then for all $\alpha \in \bar{K}$ and $\sigma \in \text{Aut}_K(\bar{K})$

$$|\sigma(\alpha)| = |\alpha|$$

Proof:

First, $\alpha, \sigma(\alpha)$ have the same minimal polynomial, for if f is the minimal polynomial of α :

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = \sigma(0) = 0$$

Thus, from number theory ([4]) we have that

$$N_{K(\alpha)/K}(\alpha) = f(0) = N_{K(\sigma(\alpha))/K}(\sigma(\alpha))$$

Then by theorem 1.4.7, setting $n = \deg f$ we get:

$$|\sigma(\alpha)| = |N_{K(\sigma(\alpha))/K}(\sigma(\alpha))|^{1/n} = |N_{K(\alpha)/K}(\alpha)|^{1/n} = |\alpha|$$

completing the proof.

Definition 1.7.2: Belonging

Let K be the field of fraction of a complete DVR with absolute value $|\cdot|$ extended to $\overline{K}$. Then for any $\alpha, \beta \in \overline{K}$, we say β *belongs* to α if

$$|\beta - \alpha| < |\beta - \sigma(\alpha)| \quad \forall \sigma \in \text{Aut}_K(\overline{K}) \setminus \{\sigma \mid \sigma(\alpha) = \alpha\}$$

In other words, β is strictly closer to α than all its conjugates.

As all nonarchimedean triangles are isosceles⁵, it is equivalent to

$$|\beta - \alpha| < |\alpha - \sigma(\alpha)|$$

Theorem 1.7.3: Krasner's Lemma

Let K be a field of fraction of a complete DVR, $\alpha, \beta \in \overline{K}$, and α is separable over K . Then if β belongs to α , $K(\alpha) \subseteq K(\beta)$ (i.e. $\alpha \in K(\beta)$).

Proof:

For the sake of contradiction, suppose β belongs to α but $\alpha \notin K(\beta)$. Then $K(\alpha, \beta)/K(\beta)$ is separable and nontrivial, thus there exists an automorphism $\sigma \in \text{Aut}_{K(\beta)}(K(\alpha, \beta))$ where $\sigma(\alpha) \neq \alpha$. Applying lemma 1.7.1 to $\beta - \alpha \in \overline{K}$:

$$|\beta - \alpha| = |\sigma(\beta - \alpha)| = |\sigma(\beta) - \sigma(\alpha)| = |\beta - \sigma(\alpha)|$$

where σ fixes α by definition. But this not contradicts that β belongs to α , completing the proof.

1.7.4 Hensel-like characterization Sutherland comments how this is similar to Hensel's lemma in that it also characterizes Henselian fields, and that the lemma was proven by Ostrowski.

Example 1.7.5: Krasner's Lemma

Give an example of it failing for a non-Henselian field.

Definition 1.7.6: L1 Norm

Let K be a field with absolute value $|\cdot|$, and let $f = \sum_i f_i x^i \in k[x]$. Then

$$\|f\|_1 = \sum_i |f_i|$$

The reader should verify that this is indeed a norm on the K -vector space $K[x]$.

⁵If one side is shorter than another, it is shorter than both

Lemma 1.7.7: Roots Smaller Than Polynomial

Let K be a field with absolute value $|\cdot|$, $f = \prod_i (x_i - \alpha_i) \in K[x]$ be a monic polynomial with $\alpha_1, \dots, \alpha_n \in L$ where L/K is a field with absolute value extending $|\cdot|$. Then for every root α_i of f :

$$|\alpha_i| < \|f\|_1$$

Proof:

If $n = 1$, then:

$$|\alpha| < |\alpha| + 1 = \|f\|_1$$

So assume $n \geq 2$. If $\|f\|_1 = 1$ then $f = x^n$ so $\alpha = |\alpha| = 0$. If $\|f\|_1 > 1$, given $|\alpha| \leq 1$ the lemma still holds ($\|f\|_1 \geq 1 + |\alpha| > |\alpha|$). So consider the case $|\alpha| > 1$. Then:

$$0 = |f(\alpha)| = \left| \alpha^n + \sum_i^{n-1} f_i \alpha^i \right| \stackrel{(1)}{\geq} |\alpha|^n - \sum_i^{n-1} |f_i| |\alpha|^i \stackrel{(|\alpha| > 1)}{\geq} |\alpha|^n - |\alpha|^{n-1} \sum_i^{n-1} |f_i| \stackrel{(2)}{\geq} |\alpha| - (\|f\|_1 - 1)$$

where $\stackrel{(1)}{\geq}$ comes from some absolute value manipulation, namely:

$$|a| = |a + b - b| \leq |a + b| + |-b| = |a + b| + |b| \Rightarrow |a + b| \geq |a| - |b|$$

and $\stackrel{(2)}{\geq}$ comes from dividing by $|\alpha|^{n-1} \geq 1$. Re-arranging:

$$\|f\|_1 - 1 \geq |\alpha| \quad \text{i.e.} \quad \|f\|_1 \geq |\alpha| + 1 > |\alpha|$$

as we sought to show.

Theorem 1.7.8: Continuity of Roots

Let K be a field of fraction of a complete DVR and $f \in K[x]$ a monic irreducible separable polynomial. Then there exists a $\delta = \delta \in \mathbb{R}_{>0}$ such that for every monic polynomial $g \in K[x]$ where $\|f - g\|_1 < \delta$,

every root β of g belongs to a root α of f for which $K(\alpha) = K(\beta)$

In particular, every such g is irreducible, separable, and has the same splitting field as f

Proof:

First, given the theorem is true and δ gives the desired result, then we can choose $\delta' = \min\{1, \delta\}$ so then any monic polynomial g satisfying $\|f - g\|_1 < \delta \leq 1$ is forced to have the same degree as f ; thus we can assume $\deg f = \deg g$.

To prove the theorem, first fixed an algebraic closure $\overline{K}$ and extend $|\cdot|$. Let $\alpha_1, \dots, \alpha_n$ be the roots of f in $\overline{K}$ so that:

$$f = \prod_i^n (x - \alpha_i) = \sum_i^n f_i x^i$$

Next, let

$$\epsilon = \min_{\alpha_i \neq \alpha_j} \{1, |\alpha_i - \alpha_j|\}$$

Then define δ with respect to f to be:

$$\delta = \left(\frac{\epsilon}{2(|f|_1 + 1)} \right)^n > 0$$

As f is monic, $|f|_1 \geq 1$ so as $\epsilon \leq 1$, $\delta < 1$. Now suppose $g = \sum_i^n g_i x^i$ is a monic polynomial st $|f - g|_1 < \delta$. Then by adding $f - f$ and the triangle inequality

$$|g|_1 \leq |f|_1 + |f - g|_1 < |f|_1 + \delta$$

Furthermore, for any root β of g :

$$|f(\beta)| = |f(\beta) - 0| = |f(\beta) - g(\beta)| = |(f - g)(\beta)| = \left| \sum_i^n (f_i - g_i) \beta^i \right| \leq \sum_i^n |f_i - g_i| |\beta|^i$$

To bound this further, by lemma 1.7.7 $|\beta| < |g|_1$ and as $|g|_1 \geq 1$, $|\beta|^i < |g|_1^{-i} \leq |g|_1^n$. Thus:

$$|f(\beta)| < |f - g|_1 |g|_1^n < \delta(|f|_1 + \delta)^n < \delta(|f|_1^n + 1)^n \leq \left(\frac{\epsilon}{2} \right)^n$$

Squishing these inequalities, we get:

$$\prod_i^n |\beta - \alpha_i| = |f(\beta)| < \left(\frac{\epsilon}{2} \right)^n$$

Then it must be that for at least one α_i :

$$|\beta - \alpha_i| < \frac{\epsilon}{2}$$

In fact, this α_i is the unique root that has this property, for by the triangle inequality

$$|\alpha_i - \alpha_j| \geq \epsilon \quad i \neq j$$

Thus, letting $\alpha = \alpha_i$, β belongs to α . By Krasner's lemma (theorem 1.7.3), $K(\alpha) \subseteq K(\beta)$, and as $n \leq [K(\alpha) : K] \leq [K(\beta) : K] \leq n$ we in fact have $K(\alpha) = K(\beta)$. Then as $n = \deg(g) = [K(\beta) : K]$, g must be the minimal polynomial (and hence irreducible) for β . Furthermore, as $K(\alpha)$ is separable, g must be separable.

To show they have the same splitting field, note that if a root β of g belongs to a root α of f , then for any $\sigma \in \text{Aut}_K(\overline{K})$ such that $\sigma(\alpha) \neq \alpha$:

$$|\tau(\beta) - \tau(\alpha)| = |\tau(\alpha - \beta)| = |\beta - \alpha| < |\alpha - \sigma(\alpha)| = |\tau(\alpha - \sigma(\alpha))| = |\tau(\alpha) - \tau(\sigma(\alpha))|$$

(note that $\sigma(\alpha) \neq \alpha \Leftrightarrow \tau(\sigma(\alpha)) \neq \tau(\alpha)$). Thus, as β belongs to α , $\tau(\beta)$ belongs to $\tau(\alpha)$. As $\text{Aut}_K(\overline{K})$ acts transitively on the roots of f and g , every root β of g belongs to a dissent root α of f , and so $K(\alpha) = K(\beta)$ each root. It follows that f, g have the same splitting field, completing the proof.

1.8 Local-Global Correspondences

Let $\hat{L}$ be a local field. Then either by definition or by proposition 1.2.2 we know that $\hat{L}$ is a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ for some $p \leq \infty$ or some $q = p^r$. Furthermore, any completion of a global field with any nontrivial absolute value is a local field. It is thus reasonable to ask if we can complete the square; where $\hat{L}$ is the completion of a corresponding global field L that is a finite extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$. In the local number field case:

$$\begin{array}{ccc} \mathbb{Q}_p & \xrightarrow{\text{finite ext.}} & \hat{L} \\ \uparrow | \cdot |_p \text{ compl.} & & \uparrow | \cdot |_p \text{ compl.} \\ \mathbb{Q} & \xrightarrow{\text{finite ext.}} & L \end{array}$$

Generalizing the question, given a fixed global field K and a local field $\hat{K}$ that is the completion of K with respect to a nontrivial absolute value $|\cdot|$, if it the case that every finite extension of local field $\hat{L}/\hat{K}$ corresponds to an extension of global field L/K where $\hat{L}$ is the completion of L with respect to to an absolute value whose restriction to K must be equivalent to $|\cdot|$. This is indeed the case! We shall show this for the separable case as it is what we need and the inseparable case offers more complication

Theorem 1.8.1: Local to Global Correspondence

Let K be a global with with a non-trivial absolute value $|\cdot|$ and $\hat{K}$ the completion of K with respect to $|\cdot|$. Then every finite separable extension $\hat{L}/\hat{K}$ is the completion of a finite separable extension of L of K with respect to an absolute value that restricts to $|\cdot|$. Furthermore, L can be chosen so that

$$[L : K] = [\hat{L} : \hat{K}]$$

in which case $\hat{L} = \hat{K} \cdot L$.

Proof:

Let $\hat{L}/\hat{K}$ be a separable extension of degree n . Let K have an absolute value. If $|\cdot|$ is Archimedean, then K is a number field and $\hat{K} \in \{\mathbb{R}, \mathbb{C}\}$, in which case these cases are left to the reader.

So assume $|\cdot|$ is nonarchimedean so that the valuation ring of $\hat{K}$ is a complete DVR and $|\cdot|$ is induced by its discrete valuation. As the extension is separable, by the primitive element theorem, we have a monic irreducible and separable polynomial $f \in \hat{K}[x]$ such that

$$\hat{L} = \frac{\hat{K}[x]}{f}$$

Let us now find an appropriate $g \in K[x]$. As K is dense in its completion $\hat{K}$, we can find a monic $g \in K[x]$ such that $\|g - f\|_1 < \delta$ for any $\delta > 0$. Then by theorem 1.7.8, g is separable and

$$\hat{L} = \frac{\hat{K}[x]}{g}$$

Then $\hat{L}$ is a finite separable extension of the field of fraction of a complete DVR, and so it is a field of fraction of a complete DVR and has a unique absolute value that extends the absolute value $|\cdot|$ on $\hat{K}$.

Now, consider $L := K[x]/(g)$. The polynomial is irreducible in $\hat{K}[x]$, and hence in $K[x]$, and so

$$[L : K] = \deg g = [\hat{L} : \hat{K}]$$

Note the field $\hat{L}$ contains both $\hat{K}$ and L , and by construction is the smallest field that does this (note that g is irreducible in $\hat{K}[x]$), and so

$$\hat{L} = \hat{K}L$$

Finally, the absolute value on $\hat{L}$ restricts to an absolute value on L extending the absolute value $|\cdot|$ on K , and hence $\hat{L}$ is complete and hence contains the completion of L with respect to $|\cdot|$. On the other hand, the completion of L with respect to $|\cdot|$ contains L and $\bar{K}$, and so must be $\hat{L}$, as we sought to show.

Let us now assume that $\hat{L}/\hat{K}$ is galois. Then is the corresponding global extension L/K galois, and are their galois groups isomorphic? As it turns out, it need not even be galois:

Example 1.8.2: Corresponding Global Extension Not Galois

Let $K = \mathbb{Q}$ and $\hat{K} = \mathbb{Q}_7$. Take $\hat{L} = \hat{K}[x]/(x^3 - 2)$. This extension is galois. The extension by definition contains the cube-root of 2, $\sqrt[3]{2}$. The other roots are this root times ζ_3 , which is contains in $\mathbb{Q}_7$. This is true as $x^2 + x + 1 \in \mathbb{F}_7[x]$ has two roots which by Hensel's lemma both lift to roots in $\mathbb{Q}_7[x]$. However, $L = K[x]/(x^3 - 2)$ is *not* a galois extension, containing only one root.

However, if we replace K with $\mathbb{Q}(\zeta_3)$, this doesn't change the completion to $\hat{K}$ (take the completion of K with respect to the absolute value induced by a prime above 7), and certainly doesn't change $\hat{L}$, but now $L = K[x]/(x^3 - 2)$ is a galois extension.

Is it always possible to give a nice enough adjustment in the base global field K to get a correspondence of galois groups? This turns out to be the case.

Corollary 1.8.3: Local To Global Correspondence, Galois Groups

Let $\hat{L}/\hat{K}$ be a finite galois extension of local fields. Then there exists a finite galois extension of global fields L/K and an absolute value $|\cdot|$ on L such that $\hat{L}$ is the completion of L with respect to $|\cdot|$, $\hat{K}$ is the completion of K with respect to the restriction of $|\cdot|$ to K , and

$$\text{Gal}_K(L) \cong \text{Gal}_{\hat{K}}(\hat{L})$$

Proof:

The Archimedean case is immediate so assume $|\cdot|$ is nonarchimedean on $\hat{L}$ and it restricts to an absolute value on $\hat{K}$. By theorem 1.8.1 we may assume that this field is the completion of a global field K with respect to the restriction of $|\cdot|$. As in the proof of that theorem, take a monic separable irreducible polynomial $g \in K[x]$ that has all those properties in $\hat{K}[x]$ such that $\hat{L} = \hat{K}[x]/(g)$ and define again $L = K[x]/(g)$ so that $\hat{L} = \hat{K}L$.

This time, let M/K be the splitting field of g . As $\hat{L}/\hat{K}$ is galois, $\hat{L}$ contains the roots too, and so by the universal property of splitting fields $\hat{L}$ contains a subextension in of K isomorphic to M ; we may identify this extension with M .

Now, note that $\hat{L}$ is equal to the completion of M with respect to to the restriction of $|\cdot|$ to M . From this we have a group homomorphism induce by restriction:

$$\varphi : \text{Gal}_{\hat{K}}(\hat{L}) \rightarrow \text{Gal}_K(M)$$

This map is injective, as each $\sigma \in \text{Gal}_{\hat{K}}(\hat{L})$ is determined by its action on the roots of g in M . Now, replace K by a fixed field of the image of φ , and replace L with M . The completion of K with respect to to the restriction of $|\cdot|$ is still $\hat{K}$, and similarity for L and $\hat{L}$, and so:

$$\text{Gal}_{\hat{K}}(\hat{L}) \cong \text{Gal}_K(L)$$

as we sought to show.

Let us now put this all together and consider the $AKLB$ case and look a primes $\mathfrak{P}$ lying over $\mathfrak{p}$:

Theorem 1.8.4: Global To Local Correspondence

Assume $AKLB^a$, $\mathfrak{p} \subseteq A$, $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}}}$. Let $K_{\mathfrak{p}}$ be the completion of K with respect to $|\cdot|_{\mathfrak{p}}$ which has maximal ideal $\hat{\mathfrak{p}}$ in its valuation ring. Similarly, for each $\mathfrak{P}/\mathfrak{p}$, let $L_{\mathfrak{P}}$ denote the completion of L with respect to $|\cdot|_{\mathfrak{P}}$ and let $\hat{\mathfrak{P}}$ denote the maximal ideal of the corresponding maximal ideal of the valuation ring. Then:

1. Each $L_{\mathfrak{P}}$ is a finite separable extension of $K_{\mathfrak{p}}$ with $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] \leq [L : K]$.
2. Each $\hat{\mathfrak{P}}$ is the unique prime of $L_{\mathfrak{P}}$ lying over $\hat{\mathfrak{p}}$.
3. Each $\hat{\mathfrak{P}}$ has ramification index $e_{\hat{\mathfrak{P}}} = e_{\mathfrak{P}}$ and residue field degree $f_{\hat{\mathfrak{P}}} = f_{\mathfrak{P}}$.
4. $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] = e_{\mathfrak{P}} f_{\mathfrak{P}}$
5. The map $L \otimes_K K_{\mathfrak{p}} \rightarrow \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$ given by $\ell \otimes x \mapsto (\ell x, \dots, \ell x)$ is an isomorphism of finite étale $K_{\mathfrak{p}}$ algebras
6. If L/K is galois, then each $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is galois and we have isomorphisms of decomposition groups and inertia groups:

$$D_{\mathfrak{P}} \cong D_{\hat{\mathfrak{P}}} = \text{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}}) \quad I_{\mathfrak{P}} \cong I_{\hat{\mathfrak{P}}}$$

^aRecall L/K is finite and separable

Proof:

(2) follows from theorem 1.4.4, (3) from simple computations, (4) from (2) and (3) and the fundamental identity. We'll show 1, 5, and 6

1. For each $\mathfrak{P}/\mathfrak{p}$, the embedding $K \hookrightarrow L$ induces an embedding $K_{\mathfrak{p}} \rightarrow L_{\mathfrak{P}}$ by mapping the equivalence classes, which is well-defined as a sequence which is Cauchy in K with respect to $|\cdot|_{\mathfrak{p}}$ is also Cauchy in L with respect to $|\cdot|_{\mathfrak{P}}$ (the valuation is extended). Thus $K_{\mathfrak{p}}$

can be viewed as a topological subfield of $L_{\mathfrak{P}}$. To show $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] \leq [L : K]$, first any K -basis $b_1, \dots, b_m$ for $L \subseteq L_{\mathfrak{P}}$ spans $L_{\mathfrak{P}}$ as a $K_{\mathfrak{p}}$ -vector space. To see this, if $y = (y_n)$ is a Cauchy sequence of elements in L , then we can write for each y_n position:

$$y_n = x_{1,n}b_1 + \dots + x_{m,n}b_m \quad x_{i,n} \in K$$

these all will give Cauchy-sequences, as linear maps of finite dimensional normed space are uniformly continuous, and thus preserve Cauchy sequences.

Next, L is a finite étale K -algebra as L/K is separable. Thus the change of basis $L \otimes_K K_{\mathfrak{p}}$ to $K_{\mathfrak{p}}$ is a finite étale $K_{\mathfrak{p}}$ -algebra (see [4]). Now, consider the $K_{\mathfrak{p}}$ -algebra homomorphism:

$$\varphi_{\mathfrak{P}} : L \otimes_K K_{\mathfrak{p}} \rightarrow L_{\mathfrak{P}} \quad \ell \otimes x \mapsto \ell x$$

We have that $\varphi_{\mathfrak{P}}(b_i \otimes 1) = b_i$ for each K -basis element $b_i \in L$, and as $b_1, \dots, b_m$ span $L_{\mathfrak{P}}$ as a $K_{\mathfrak{p}}$ -vector space, this map is surjective. Next, $L \otimes_K K_{\mathfrak{p}}$ is by definition finite étale algebraic, hence it is isomorphic to a finite product of finite separable extensions of $K_{\mathfrak{p}}$. But then $L_{\mathfrak{P}}$ is isomorphic to a subproduct, and thus is also a finite étale $K_{\mathfrak{p}}$ algebra, implying $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is separable and finite.

5. From (1), combine each $\varphi_{\mathfrak{P}}$ to define the $K_{\mathfrak{p}}$ -algebra homomorphism:

$$\varphi = \prod_{\mathfrak{P}/\mathfrak{p}} \varphi_{\mathfrak{P}} \quad \varphi : L \otimes_K K_{\mathfrak{p}} \rightarrow \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$$

As dimension of étale algebras is base-change invariant (again see [4]):

$$\dim_{K_{\mathfrak{p}}}(L \otimes_K K_{\mathfrak{p}}) \dim_K L = [L : K] = \sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}} f_{\mathfrak{P}} = \sum_{\mathfrak{P}/\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = \dim_{K_{\mathfrak{p}}} \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$$

To show φ is an isomorphism, it suffices to show the map is surjective (as the dimensions line up). If we can pick a $K_{\mathfrak{p}}$ -basis $\mathcal{B}$ for $L \otimes_K K_{\mathfrak{p}}$ and show that $\det([\varphi]_{\mathcal{B}}) \neq 0$, the result is proven. To that end, pick a basis $\mathcal{B}$ of tuples $\beta_i = (\beta_{i,\mathfrak{P}})_{\mathfrak{P}/\mathfrak{p}}$ for the codomain. To find a basis that maps to it, by theorem 1.7.8 pick $\alpha_i \in L$ such that

$$|\alpha_i - \beta_{i,\mathfrak{P}}|_{\mathfrak{P}} < \epsilon \quad \forall \mathfrak{P}/\mathfrak{p}$$

Thus, given the product the sup-norm, each $\varphi(\alpha_i \otimes 1)$ is “close” to β_i ; in particular the matrix M where each column expresses $\varphi(\alpha_j \otimes 1)$ in terms of β_i is closed to the identity matrix (note the determinant is continuous!), and hence the determinant is close to 1. Then for ϵ small enough, it must be that $\det(M) \neq 0$, and so φ is surjective, and thus an isomorphism.

6. Assume L/K is galois. Then each $\sigma \in D_{\mathfrak{P}}$ that acts on L respect $v_{\mathfrak{P}}$ as it fixes $\mathfrak{P}$, and so σ preserves Cauchy-sequences, showing that σ is an automorphism of $L_{\mathfrak{P}}$ and fixes $K_{\mathfrak{p}}$. We thus get a well-defined map:

$$\varphi : D_{\mathfrak{P}} \rightarrow \text{Aut}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}})$$

This map is injective, for if σ acts trivially on $L_{\mathfrak{P}}$, certainly acts trivially on L . For surjectivity, note that :

$$e_{\mathfrak{P}} f_{\mathfrak{P}} = |D_{\mathfrak{P}}| \leq \#\text{Aut}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}}) \leq [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = e_{\mathfrak{P}} f_{\mathfrak{P}}$$

and so $\#\text{Aut}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}}) \leq [L_{\mathfrak{P}} : K_{\mathfrak{p}}]$ showing that $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is galois and that φ is surjective, and hence an automorphism. Next, as $\hat{\mathfrak{P}}$ is the only prime of $L_{\mathfrak{P}}$, it must be fixed by every $\sigma \in \text{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}})$ and so

$$D_{\mathfrak{P}} \cong D_{\hat{\mathfrak{P}}} = \text{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}})$$

Finally, the inertia groups $I_{\mathfrak{P}}$ and $\hat{I}_{\mathfrak{P}}$ both have the same order and φ restricts to an (injective) homomorphism between these groups, and hence they are isomorphic.

Corollary 1.8.5: Global To Local Correspondence, Norm and Trace

Assume AKLB and let $\mathfrak{p}$ be a prime of A . Then for every $\alpha \in L$, we have:

$$N_{L/K}(\alpha) = \prod_{\mathfrak{P}/\mathfrak{p}} N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(\alpha) \quad \text{tr}_{L/K}(\alpha) = \sum_{\mathfrak{P}/\mathfrak{p}} \text{tr}_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(\alpha)$$

Proof:

The norm and trace defined by the determinant and trace of the K -linear map $L \xrightarrow{\times \alpha} L$ is unchained when tensoring with $K_{\mathfrak{p}}$, hence by the isomorphism in part (5) of the above theorem we get the desired result.

Corollary 1.8.6: Global To Local Correspondence, Decomposition

Assume AKLB, $\mathfrak{p}$ a prime of A , $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}}}$, $\hat{A}_{\mathfrak{p}}$ the completion of A with respect to $|\cdot|_{\mathfrak{p}}$ and for each $\mathfrak{P}/\mathfrak{p}$ let $\hat{B}_{\mathfrak{P}}$ denote the completion of B with respect to $|\cdot|_{\mathfrak{P}}$. Then

$$B \otimes_A \hat{A}_{\mathfrak{p}} \cong_{\hat{A}_{\mathfrak{p}}} \prod_{\mathfrak{P}/\mathfrak{p}} \hat{B}_{\mathfrak{P}}$$

Proof:

Choose some prime $\mathfrak{p} \subseteq A$. Then $B_{\mathfrak{p}}$ is a free A -module of rank $n = [L : K] = \sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}} f_{\mathfrak{P}}$, so $B \otimes_A \hat{A}_{\mathfrak{p}}$ is a free $\hat{A}_{\mathfrak{p}}$ -module of rank n . Next, as $\hat{A}_{\mathfrak{p}}$ and $\hat{B}_{\mathfrak{P}}$ are DVR's with respect to their respective primes, by theorem 1.8.4(4) $\prod_{\mathfrak{P}/\mathfrak{p}} B_{\mathfrak{P}}$ is a free $\hat{A}_{\mathfrak{p}}$ -module of rank n . Finally, by theorem 1.8.4 these two $\hat{A}_{\mathfrak{p}}$ -modules both lie in isomorphic finite étale $K_{\mathfrak{p}}$ -algebras $L \otimes_K K_{\mathfrak{p}} \cong \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$, and it is clear this $\hat{K}_{\mathfrak{p}}$ -isomorphism restricts to a $\hat{A}_{\mathfrak{p}}$ -isomorphism, completing the proof.

Put some stuff about Different and Discriminant from Sutherland that make sense; the rest is in EYTNA Algebra

1.9 Kronecker-Weber Theorem

We are now in a position to prove the famous Kronecker-Weber Theorem using local fields. Let $K/\mathbb{Q}$ be a finite abelian extension so that $\text{Gal}_{\mathbb{Q}}(K) = G$. Then we know that G can be represented as:

$$G \cong (\mathbb{Z}/r_1\mathbb{Z})^{n_1} \times \cdots \times (\mathbb{Z}/r_k\mathbb{Z})^{n_k}$$

We can embed G into $\mathbb{C}$ with some clever multiplication tricks. The question becomes if we can do the same with the field extensions:

Theorem 1.9.1: Global Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}$ lies in a Cyclotomic field $\mathbb{Q}(\zeta_m)$

Proof:
soon

This result can be readily deduced from the local result:

Theorem 1.9.2: Local Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}_p$ lies in a Cyclotomic field $\mathbb{Q}_p(\zeta_m)$

Proof:
soon

Proposition 1.9.3: Local Implies Global Kronecker-Weber

The Local Kronecker-Weber Theorem implies the Global Kronecker-Weber Theorem

Proof:

Let $K/\mathbb{Q}$ be a finite abelian extension. Then for each ramified prime p of $\mathbb{Q}$, pick a prime $\mathfrak{p}/p$ and let $K_{\mathfrak{p}}$ be the completion of K at $\mathfrak{p}$. As $K/\mathbb{Q}$ is galois, it makes no difference which $\mathfrak{p}$ is picked. Then we know by theorem 1.8.4 that:

$$\text{Gal}_{\mathbb{Q}_p}(K_{\mathfrak{p}}) \cong D_{\mathfrak{p}} \subseteq \text{Gal}_{\mathbb{Q}}(K)$$

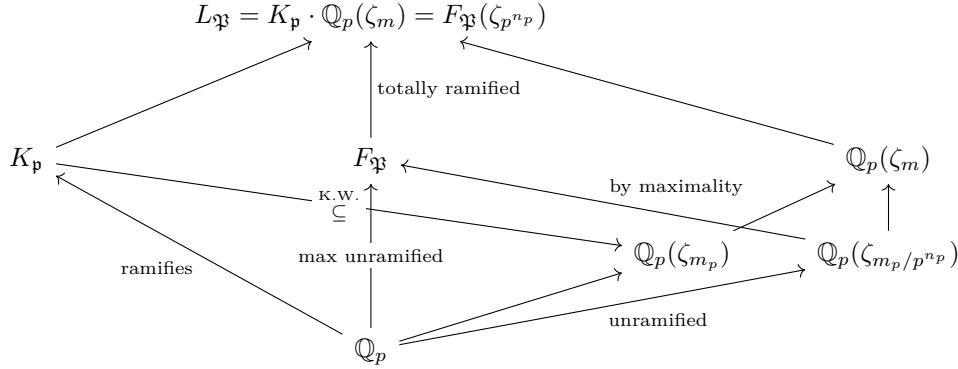
Hence $K_{\mathfrak{p}}$ is an abelian extension of $\mathbb{Q}_p$. By the local Kronecker-Weber Theorem we have that $K_{\mathfrak{p}} \subseteq \mathbb{Q}_p(\zeta_{m_p})$ for some $m_p \in N_{>0}$. Now, let $n_p = \nu_p(m_p)$, and define:

$$m = \prod_p p^{n_p}$$

which is finite as there are only finitely many ramified primes. Define $L = K(\zeta_m)$. Then we will show that $L = \mathbb{Q}(\zeta_m)$, showing that $K \subseteq \mathbb{Q}(\zeta_m)$.

First, the field $L = K \cdot \mathbb{Q}(\zeta_m)$ is the compositum of galois extensions of $\mathbb{Q}$, and hence is galois over $\mathbb{Q}$ with galois group isomorphic to a subgroup of $\text{Gal}_{\mathbb{Q}}(K) \times \text{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_m))$, and so is abelian. Let $\mathfrak{P}$ be a prime of L lying above the ramified prime $\mathfrak{p}/p$. Then the completion $L_{\mathfrak{P}}$ is a finite

abelian extension of $\mathbb{Q}_p$ as $L/\mathbb{Q}$ is finite abelian. By minimality, $L_{\mathfrak{p}} = K_{\mathfrak{p}} \cdot \mathbb{Q}_p(\zeta_m)$. Now, let $F_{\mathfrak{p}}$ be the maximal unramified extension of $\mathbb{Q}_p$ in $L_{\mathfrak{p}}$. Then $L_{\mathfrak{p}}/F_{\mathfrak{p}}$ is totally ramified, giving $\text{Gal}_{F_{\mathfrak{p}}}(L_{\mathfrak{p}}) \cong I_{\mathfrak{p}} = I_p$ (by theorem 1.8.4). Now, since $K_{\mathfrak{p}} \subseteq \mathbb{Q}_p(\zeta_{m_p})$ and $\mathbb{Q}_p(\zeta_{m_p}/p^{n_p})$ is unramified by lemma 1.6.5, we have $L_{\mathfrak{p}} = F_{\mathfrak{p}}(\zeta_{p^{n_p}})$ (since $\mathbb{Q}_p(\zeta_{p^{n_p}})$ is unramified). Hence $K_{\mathfrak{p}} \subseteq F_{\mathfrak{p}}(\zeta_{p^{n_p}})$. There are a lot of terms going around, so we may visualize them in the following diagram:



Furthermore, $F_{\mathfrak{p}} \cap \mathbb{Q}_p(\zeta_{p^{n_p}}) = \mathbb{Q}_p$ as $\mathbb{Q}_p(\zeta_{p^{n_p}})$ is totally ramified, hence:

$$I_p \cong \text{Gal}_{F_{\mathfrak{p}}}(L_{\mathfrak{p}}) \cong \text{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p(\zeta_{p^{n_p}}) \cong (\mathbb{Z}/p^{n_p}\mathbb{Z})^*$$

Now, let I be the group generated by the union of the groups $I_p \subseteq \text{Gal}_{\mathbb{Q}}(L)$ for $p|m$. Then since $\text{Gal}_{\mathbb{Q}}(L)$ is abelian, $I \subseteq \prod_p I_p$, and so:

$$\#I \leq \prod_{p|m} \#I_p = \prod_{p|m} \#(\mathbb{Z}/p^{n_p}\mathbb{Z}) = \prod_{p|m} \varphi(p^{n_p}) = \varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

Now, recall that each inertia field L^I_p is unramified at p , and $L^I \subseteq L^{I_p}$, and so $L^I/\mathbb{Q}$ is unramified, but then by definition of L^I , $L^I = \mathbb{Q}$. Then:

$$[L : \mathbb{Q}] = [L : L^I] = \#I \leq [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

giving $\mathbb{Q}(\zeta_m) \subseteq L$, which must mean that $L = \mathbb{Q}(\zeta_m)$, but then $K \subseteq L = \mathbb{Q}(\zeta_m)$, as we sought to show.

Before proving the main result, we shall prove a quick important lemma:

Lemma 1.9.4: Kronecker Weber Theorem, Lemma

For any prime p ,

$$\mathbb{Q}_p \left((-p)^{1/(p-1)} \right) = \mathbb{Q}_p(\zeta_p)$$

Proof:

Let $\alpha = (-p)^{1/(p-1)}$. Then α is the root of the Eisenstein polynomial $x^{p-1} + p$, and hence the extension $\mathbb{Q}_p(\alpha)$ is totally ramified of degree $p-1$ and α is a uniformizer. Let $\pi = \zeta_p - 1$. The minimal polynomial of π is:

$$f(x) = \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \cdots + p$$

which is Eisenstein, so $\mathbb{Q}_p(\pi) = \mathbb{Q}_p(\zeta_p)$ and is totally ramified of degree $p-1$ with π as a uniformizer. Then we have $u = -\pi^{p-1}/p \equiv 1 \pmod{\pi}$ so u is a unit in the ring of integers of $\mathbb{Q}_p(\zeta_p)$. Setting $g(x) = x^{p-1} - u$ then $g(1) \equiv 0 \pmod{\pi}$ and $g'(1) = p-1 \not\equiv 0 \pmod{\pi}$, and so by Hensel's lemma we can lift 1 to a root β of $g(x)$ in $\mathbb{Q}_p(\zeta_p)$.

We now have $p\beta^{p-1} = pu = -\pi^{-1}$, hence $(\pi/\beta)^{p-1} + p = 0$, and so $\pi/\beta \in \mathbb{Q}_p(\zeta_p)$ is a root of the minimal polynomial of α . As $\mathbb{Q}_p(\zeta_p)$ is galois, $\alpha \in \mathbb{Q}_p(\zeta_p)$. And since both $\mathbb{Q}_p(\alpha)$ and $\mathbb{Q}_p(\zeta_p)$ both have degree $p-1$, the fields are equal, as we sought to show.

Proof:**of the local Kronecker Weber Theorem**

Let us first reduce the proof to cyclic extensions of prime-power degree. This comes down to the correspondence between galois groups and compositum's. Namely, if $L_1/K, L_2/K$ are extensions where $L_1 \cap L_2 = K$, then their galois group is the product of the two galois groups. Then by the classification of finite abelian groups, we may decompose the abelian extension L/K into a compositum $L = L_1 \cdots L_n$ of linearly disjoint cyclic extensions L_i/K of prime-power degree. Then if each L_i lies in a Cyclotomic extension $K(\zeta_{m_i})$, so does L since

$$L \subseteq K(\zeta_{m_1}) \cdots K(\zeta_{m_n}) = K(\zeta_m) \quad m = m_1 \cdots m_n$$

Hence, we shall focus on cyclic extensions $K/\mathbb{Q}_p$ of prime power l^r . This breaks up into two cases: $\ell = p$ and $\ell \neq p$.

The $\ell \neq p$ case

Let F be the maximal unramified extension of $\mathbb{Q}_p$ in K . Then by proposition 1.6.6 we have $F = \mathbb{Q}_p(\zeta_n)$. Then K/F is totally ramified, and as $\ell \neq p$ it must be tamely ramified. Then by theorem 1.6.21 we have

$$K = F(\pi^{1/e})$$

for some uniformizer π with $e = [K : F]$. Take $\pi = -pu$ for some $u \in \mathcal{O}_F^\times$, as $F/\mathbb{Q}_p$ is unramified^a. Then the field $K = F(\pi^{1/e}) \subseteq F((-p)^{1/e}) \cdot F(u^{1/e})$. We shall show both lie in a Cyclotomic extension of $\mathbb{Q}_p$.

Now, since $\nu_{\mathfrak{p}}(\text{disc}(x^e - u)) = 0$ for $p \nmid e$, $F(u^{1/e})/F$ is unramified, so $F(u^{1/e})/\mathbb{Q}_p$ is unramified, hence $F(u^{1/e}) = \mathbb{Q}_p(\zeta_k)$ for some $k \in \mathbb{N}_{>0}$. Take the compositum $K(u^{1/e}) = K\mathbb{Q}_p(\zeta_k)$, which is abelian and contains the subextension $\mathbb{Q}_p((-p)^{1/e})/\mathbb{Q}_p$ which is galois as it lies in an abelian extension. It is totally ramified since it is an Eisenstein extension. Next, the field $\mathbb{Q}_p((-p)^{1/e})$ contained ζ_e by taking the ratios of roots of $x^e + p$, but $\mathbb{Q}_p(\zeta_e)/\mathbb{Q}_p$ is unramified as $p \nmid e$, so it must be that

$$\mathbb{Q}_p(\zeta_e) = \mathbb{Q}_p$$

Thus, $e \mid (p-1)$. By lemma 1.9.4:

$$\mathbb{Q}_p((-p)^{2/r}) \subseteq \mathbb{Q}_p((-p)^{1/(p-1)}) = \mathbb{Q}_p(\zeta_p)$$

Then $F((-p)^{1/e}) = F \cdot \mathbb{Q}_p((-p)^{1/e}) \subseteq \mathbb{Q}_p(\zeta_n)\mathbb{Q}_p(\zeta_p) \subseteq \mathbb{Q}_p(\zeta_{np})$. Hence:

$$K \subseteq F(u^{1/e}) \cdot F((-p)^{1/e}) \subseteq \mathbb{Q}(\zeta_k) \cdot \mathbb{Q}(\zeta_{np}) \subseteq \mathbb{Q}(\zeta_{knp})$$

completing the $\ell \neq p$ case.

The $\ell = p > 2$ case

Let us start with a cyclic extension $K/\mathbb{Q}_p$ of odd degree p^r . Then there are two candidates:

1. $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$, which is unramified of degree p^r
2. the index $p-1$ subfield of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ which is a totally ramified extension of degree p^r

If K is contained in the compositum of these two fields, we are done. Otherwise, the field $K(\zeta_m)$ is a galois extension of $\mathbb{Q}_p$ with:

$$\text{Gal}_{\mathbb{Q}_p}(K(\zeta_m)) \cong \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^s\mathbb{Z}$$

for some $s > 0$, in particular the first factor is from the galois group of $\mathbb{Q}_p(\zeta_{p^{p^r}-1})$, the second two from the factors from the galois group of $\mathbb{Q}_p(\zeta_{p^{r+1}})$ (note that $\mathbb{Q}_p(\zeta_{p^{p^r}-1}) \cap \mathbb{Q}_p(\zeta_{p^{r+1}}) = \mathbb{Q}_p$) and the last factor comes from the fact that by assumption $K \not\subseteq \mathbb{Q}_p(\zeta_m)$ so $\text{Gal}_{\mathbb{Q}_p(\zeta_m)}(K(\zeta_m))$ is nontrivial and must have order p^s for some $1 \leq s \leq r$.

Now, the abelian group $\text{Gal}_{\mathbb{Q}_p}(K(\zeta_m))$ has a quotient isomorphic to $(\mathbb{Z}/p\mathbb{Z})^3$, and the subfield of $K(\zeta_m)$ corresponding to this quotient is an abelian extension of $\mathbb{Q}_p$ with galois group $(\mathbb{Z}/p\mathbb{Z})^3$. I claim there is no such fields.

First, show that every totally wildly ramified galois extension of $\mathbb{Q}_p$ is cyclic for every odd p . Now, if $G = \text{Gal}_{\mathbb{Q}_p}(K) \cong (\mathbb{Z}/p\mathbb{Z})^3$, then write G as an internal direct sum of the inertia subgroup $I \leq G$ and a cyclic subgroup $H \leq G$ (as L^i is unramified), and hence a cyclic extension of $\mathbb{Q}_p$ with galois group isomorphic to $G/I \cong H$. But then L^H is totally wildly ramified abelian extension of $\mathbb{Q}_p$ whose galois group G/H is *not* cyclic, which completes the proof for the $\ell = p > 2$ case.

The $\ell = p = 2$ case

We must isolate the $p = 2$ case as there is an extension of $\mathbb{Q}_2$ with galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$, namely the Cyclotomic field $\mathbb{Q}_2(\zeta_{24}) = \mathbb{Q}_2(\zeta_3) \cdot \mathbb{Q}_2(\zeta_8)$. We apply the following analogous argument

First, the unramified Cyclotomic field $\mathbb{Q}_2(\zeta_{2^{2^r}-1})$ has galois group $\mathbb{Z}/2^r\mathbb{Z}$ and the totally ramified Cyclotomic field $\mathbb{Q}_2(\zeta_{2^{r+2}})$ has galois group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^r\mathbb{Z}$ (up to isomorphic). Let $m = (2^{2^r} - 1)(2^{r+2})$. Now, if $K \not\subseteq \mathbb{Q}_2(\zeta_m)$, we must have:

$$\text{Gal}_{\mathbb{Q}_2}(K(\zeta_m)) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} & 1 \leq s \leq r \\ (\mathbb{Z}/2^r\mathbb{Z})^2 \times \mathbb{Z}/2^s\mathbb{Z} & 2 \leq s \leq r \end{cases}$$

and thus must admit a quotient isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$. We shall show no extension of $\mathbb{Q}_2$ has either of these galois groups, showing that K lies in $\mathbb{Q}_2(\zeta_m)$ which will complete the proof.

To do this, first show that there are exactly 7 quadratic extensions of $\mathbb{Q}_2$, and so there is no extension of $\mathbb{Q}_2$ that has galois group $(\mathbb{Z}/2\mathbb{Z})^4$ since this groups has 15 subgroups of index 2 whose fixed fields yields 15 distinct quadratic extensions of $\mathbb{Q}_2$.

Next, there are only finitely many extensions of $\mathbb{Q}_2$ for of degree d for a fixed d , and these can be enumerated by the Eisenstein polynomials in $\mathbb{Q}_2[x]$ of degree d divign the relation given by Krasner's lemma. Then there are 59 quartic extensions of $\mathbb{Q}_2$ of which 12 are cyclic. This would be very tedious to show directly, and can be verified given this link: [LMFDB, 2-adic cyclic quadratic fields](#). Thus there is no extension of $\mathbb{Q}_2$ which has galois group $(\mathbb{Z}/4\mathbb{Z})^3$, as this group has 28 subgroups whose fixed fields gives 28 distinct cyclic quartic extensions of $\mathbb{Q}_2$, but then we have exhausted all possibilities, completing the proof.

^anamely if $\mathfrak{P}/p$ is the maximal ideal of $\mathcal{O}_F$, then the valuation $\nu_{\mathfrak{P}}$ extends ν_p with index 1 so $\nu_{\mathfrak{P}}(-pu) = \nu_p(-p) = 1$

1.10 *Hasse Principle on Conics

This section is not necessary for the rest of this book, but it demonstrates some of the usefulness of local fields. It shall quickly skim over the necessary algebraic curve theory, the reader can checkout [4, chapter 21], particularly section 6, for more details.

Using the rational root test ([4, chapter 11.3]) finding rational solutions to any degree polynomial with 1 variable is simple, and there are necessarily finitely many. Once 2 variables are introduced, the complexity of finding rational solutions increases greatly, and the degree plays an important role in compartmentalizing the complexity. Degree 1 polynomials in two variables are linear, and hence are immediately solvable using linear algebra. Degree 3 polynomials in two variables are *elliptic curves*⁶ and the complexity of finding rational solutions jumps greatly: see [2] for more details. Degree 4 and higher have only finitely many rational solutions by *Faltings's Theorem*, but as of writing this book there is no *constructive method* of finding these roots.

In the case of degree 2 polynomials with two variables, we can find all rational solutions using the *Hasse Principle*. We start by defining our terms:

Definition 1.10.1: Conic

Let

$$C : ax^2 + by^2 + cxy + dx + ey + f = 0$$

be an arbitrary equation of degree 2. Then if C is smooth, it is called a *conic*.

The case where C is not smooth is be reduced to the smooth case by normalizing $C^\nu \rightarrow C$ and showing the results hold after projecting. Naturally, projectivising creates a homogeneous polynomial in 3 variables of degree 2, and hence we can also think of a conic as:

$$C : F(x_1, x_2, x_3) + \sum f_{ij}x_i x_j = 0$$

where $f_{ij} = f_{ji}$ and $\det f_{ij} \neq 0$. In [4, chapter 21], it was left as an exercise to show that over any perfect field any conic has a liner-isomorphism making it equivalent to a curve of the form:

$$f_1x_1^2 + f_2x_2^2 + f_3x_3^2 = 0$$

for $f_i \neq 0$.

⁶or are singular elliptic curves

The main idea was that we can always diagonalize the coefficient matrix.

We can use some algebraic geometry to deduce the following:

Lemma 1.10.2: Conic – Single Solution

Let C be a degree 2 be a conic. Then C either has no rational solutions, or infinitely many

Proof:

Let $f(x, y)$ be the equation for C and assume $(a, b) \in \mathbb{Q}^2$ such that $f(a, b) = 0$. Construct the rest of the solutions by taking all rational-sloped lines passing through (a, b) . This in fact creates a bi-rational map between C and $\mathbb{P}_{\mathbb{Q}}^1$, showing that this is in fact *all* such solutions.

We shall show that we can find all rational solutions of a degree 2 curve in two variables

Theorem 1.10.3: Local-to-global For Conics

Let C be a conic over $\mathbb{Q}$. Then C has a rational point (a $\mathbb{Q}$ -solution to C) if and only if C has a $\mathbb{Q}_p$ -solution for each prime p and an $\mathbb{R}$ -solution.

Note that if a solution exists over $\mathbb{Q}$, as $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ for each p and $\mathbb{Q} \hookrightarrow \mathbb{R}$, a solution certainly exists over each local field (this also motivates why $\mathbb{Q}$ is called a global field: a solution in this field is trivially a solution over each local field). Thus, the converse will be proven. The key result in this proof shall be a clever application of Minkowski's theorem which recall here

Lemma 1.10.4: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in the Euclidean vector space V of dimension n and let X be a centrally symmetric convex subset of V . Then if

$$\text{vol}(X) > 2^n \text{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

Proof:

see [4, chapter 22.1]

Proof:

of theorem 1.10.3

Let $F(x_1, x_2, x_3)$ represent the conic C in projective coordinates and assume a nontrivial^a solution exists for each $\mathbb{Q}_p$ and $\mathbb{R}$. As mentioned earlier, any conic has a linear isomorphism to a curve of the form:

$$f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2$$

where each $f_i \neq 0$. By multiplying out the denominator we can assume each $f_i \in \mathbb{Z}$. Let us first show that we can reduce to the case where $f_1 f_2 f_3 \in \mathbb{Z}$ is square-free. For, if the product was

not square free, if $p^2 \mid f_i$, then apply $x_i \rightarrow \frac{1}{p}x_i$ until f_i is square free. If two of the factors have a p factor, send the third to p times itself. Then the whole expression can be divided by p^3 . Then the magnitude of $|f_1 f_2 f_3|$ must eventually terminate, giving the desired square-free reduction.

With this, let us setup how Minkowski's theorem will be key: we shall show there exists a lattice $\Lambda \subseteq \mathbb{Z}^3$ of index $m = 4|f_1 f_2 f_3|$ where

$$F(x_1, x_2, x_3) \equiv 0 \pmod{\Lambda}$$

Given such a lattice exists, we can define^b

$$S = \{(x_1, x_2, x_3) \in \mathbb{Q}^3 \mid |f_1|x_1^2 + |f_2|x_2^2 + |f_3|x_3^2 < m\}$$

Then:

$$V(S) > \frac{\pi}{3} 2^3 4|f_1 f_2 f_3| > 4|f_1 f_2 f_3| = m$$

so by Minkowski's theorem there exists a $(x, y, z) \in S \cap \Lambda$ such that

$$F(x, y, z) \equiv 0 \pmod{\Lambda} \quad \text{and} \quad |F(x, y, z)| \leq |f_1|x_1^2 + |f_2|x_2^2 + |f_3|x_3^2 < m$$

which implies a rational solution exists!

Thus, we reduce to showing such a Λ exists. First, by assumption there exists points $0 \neq (a_1, a_2, a_3) \in \mathbb{Q}_p^3$ such that $F(a_1, a_2, a_3) = 0$, as well as the real case. Certainly all multiples of (a_1, a_2, a_3) are also solution, hence without loss of generality we may assume $\max_i |a_i|_p = 1$. Thus, $(a_1, a_2, a_3) \in \mathbb{Z}_p^3$. Now, we split up the proof in 3 cases:

($p \neq 2$, $p \mid f_1 f_2 f_3$): Without loss of generality, say $p \mid f_1$ so that $p \nmid f_2 f_3$. Then $|f_1 a_1^2|_p < 1$. Now, if it were the case that $|a_1|_p < 1$, then by the ultra-metric inequality:

$$|f_3 a_3^2|_p = |f_1 a_1^2 + f_2 a_2^2| < 1$$

and so $|a_3|_p < 1$, but then:

$$|f_1 a_1^2|_p = |f_2 a_2^2 + f_3 a_3^2|_p < p^{-2}$$

implying $|a_1|_p < 1$ – contradicting our earlier normalization. Thus,

$$|a_2|_p = |a_3|_p = 1$$

and hence these are units in $\mathbb{Q}_p$. Then:

$$|f_2 a_2^2 + f_3 a_3^2| < 1$$

Now, dividing by a_2^2 (as it is a unit) we can deduce there is some $r_p \in \mathbb{Z}$ and f_p such that

$$f_2 + r_p^2 f_3 \equiv 0 \pmod{p}$$

From this, we can now find a solution to our equation. By requiring $x_3 \equiv r_p x_2 \pmod{p}$, we get

$$F(x_1, x_2, x_3) \equiv f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2 \equiv (f_2 r_p^2 f_3) x_2^2 \equiv 0 \pmod{p}$$

giving our first case

1. ($p = 2, p \nmid f_1 f_2 f_3$) By examining the coefficients mod 2, exactly two of the coefficients must be units, without loss of generality say a_1, a_2 . Then as $t^2 \equiv 0, 1 \pmod{4}$, it must be that:

$$f_2 + f_3 \equiv 0 \pmod{4}$$

Then by requiring

$$x_1 \equiv 0 \pmod{2} \quad \text{and} \quad x_2 \equiv x_3 \pmod{2}$$

we get $F(x_1, x_2, x_3) \equiv 0 \pmod{4}$, giving our second case.

2. ($p = 2, p \mid f_1 f_2 f_3$) Without loss of generality say $2 \mid f_1$, so $2 \nmid f_2, f_3$. Then like in the first case we get $|a_2|_2 = |a_3|_2 = 1$. When $a \in \mathbb{Z}$ is odd, $a^2 \equiv 1 \pmod{8}$. So we get the following two possibilities:

$$f_2 + f_3 \equiv 0 \pmod{8} \text{ iff } 2 \mid a_1$$

or:

$$f_1 + f_2 + f_3 \equiv 0 \pmod{8} \text{ iff } 2 \nmid a_1$$

In the first case, let $s = 0$ and in the second let $s = 1$. Then require:

$$x_2 \equiv x_3 \pmod{4} \quad \text{and} \quad x_1 \equiv s x_3 \pmod{2}$$

Then $F(x_1, x_2, x_3) \equiv 0 \pmod{8}$.

Now, define the lattice Λ in $\mathbb{Z}^3$ which satisfies these relations. Then by construction its index is

$$m = 4|f_1 f_2 f_3|$$

Hence

$$F(x_1, x_2, x_3) \equiv 0 \pmod{m}$$

But then, by Minkowski's theorem, we get a rational point that satisfies this equation, as we sought to show.

^aIn projective coordinates, $[0 : 0 : 0]$ does not exist; this would be the trivial solution

^bNote that the Minkowski's theorem was originally used with sublattices of number rings given by primes, however part of its strength in ability to generally conclude the existence of intersection points of lattices, just like is being done here

Places: Generalizing Primes

As we saw, primes uniquely correspond to discrete valuations (or equivalence classes of absolute values). From this perspective, the Archimedean prime could be thought of as the “prime at infinity”. This can in fact be made very precise, and shall even need to be taken properly into account when working looking at *ray fields* in chapter ?? . We thus dive deeper into this connection

2.1 Generalizing Primes

Definition 2.1.1: Places

Let K be a field. Then a *place* of K is an equivalence class of nontrivial absolute values on K . The set of places of K shall be denoted M_K

Recall that the completions of K are in correspondence with absolute values and are invariant under equivalent absolute values, and hence completions of K are in correspondence with places. If $|\cdot|_\nu$ is some representative absolute value of a place, then K_ν shall denote its completion. If a place is Archimedean, the field K_ν shall be called Archimedean, and nonarchimedean otherwise. If K is a global field, then the completion at a place gives a local field, which if ν is Archimedean we know that $K_\nu \cong \mathbb{R}$ or $K_\nu \cong \mathbb{C}$, and it is nonarchimedean then the absolute value of K_ν is induced by a discrete valuation, which we shall denote ν (this motivated the symbol for general fields, as the discrete valuation is independent of the choice of absolute value representative).

Example 2.1.2: Places

1. When $K = \mathbb{Q}$, this gives the familiar places given by the primes as well as ∞ .
2. When $K = \mathbb{F}_q[t]$ for $q = p^r$, the places correspond to the irreducible polynomials of $\mathbb{F}_q[t]$

along with a *nonarchimedean* place at infinity $|\cdot|_\infty = q^{\deg(-)}$.

The other places that shall be considered are extensions of these fields.

Definition 2.1.3: Types of Places

Let K be a global field and ν a place. Then:

1. $K_\nu \cong \mathbb{R}$, ν is said to be a *real place*
2. $K_\nu \cong \mathbb{C}$, ν is said to be a *complex place*
3. If $|\cdot|_\nu$ is induced by a discrete valuation $\nu_{\mathfrak{p}}$ for a prime $\mathfrak{p} \subseteq K$, then ν is a *finite place*. Otherwise it is an *infinite place*.

Every finite place must be nonarchimedean, every infinite place is Archimedean in characteristic 0 and nonarchimedean otherwise, every Archimedean place is an infinite place, but nonarchimedean may be finite or infinite (as demonstrated in example 2.1.2).

Definition 2.1.4: Extending Places

Let L/K be an extension of global fields. Then for every ω of L , any absolute value $|\cdot|_\omega$ that represents the equivalence class w restricts to an absolute value on K that represents a place ν of K independent of choice of $|\cdot|_\omega$. Then we write $w|\nu$ and say that w *extends* ν or that w *lies above* ν .

Here now comes the most important results about places that generalize the Archimedean embedding and theorem 1.8.4(5):

Theorem 2.1.5: Extending Global Ring Tensoring Local Ring

let L/K be a finite separable extension of global fields and let ν be a place of K . Then:

$$L \otimes_K K_\nu \cong \prod_{w|\nu} L_w$$

given by

$$\ell \otimes x \mapsto (\ell x, \dots, \ell x)$$

Note that this was already proven for the nonarchimedean places in theorem 1.8.4. The following proof shows how the ideas work for places in general:

Proof:

Let ν be a place on K . First off, the separable extension L/K is a finite étale K -algebra, so the base change $L \otimes_K K_\nu$ is a finite étale K_ν -algebra, and is therefore isomorphic to a finite product $\prod_{i \in I} L_i$ of finite separable extensions L_i of K_ν , each of which is a local field (any finite extension of a local field is a local field). Thus, it suffices to show that there is a bijective correspondence between the sets of local fields and localizations:

$$\{L_i : i \in I\} \quad \leftrightarrow \quad \{L_w : w|\nu\}.$$

To that end, fix an absolute value $|\cdot|_v$ on K_v . As L_i is a local field extending K_v , when ν is nonarchimedean by theorem 1.4.7 there is a unique absolute value $|\cdot|_w$ that restricts to $|\cdot|_v$; when ν is Archimedean it is immediate since then either $K_v \cong L_w$ or $K_v \cong \mathbb{R} \subseteq \mathbb{C} \cong L_w$ and the Euclidean absolute value on $\mathbb{R}$ is the restriction of the Euclidean absolute value on $\mathbb{C}$. Then the map $L \hookrightarrow L \otimes_K K_v \cong \prod_i L_i \twoheadrightarrow L_i$ allows us to view L as a subfield of each L_i ($L \subseteq L_i$), so $|\cdot|_w$ on L_i restricts to L that uniquely determines a place $w|v$. There is thus a well-defined map

$$\varphi : \{L_i : i \in I\} \rightarrow \{L_w : w|v\}$$

Then φ must be shown to be bijective and $\varphi(L_i) \cong L_i$.

For $\varphi(L_i) \cong L_i$, note that $L \otimes_K K_v \cong \prod_i L_i$ as an isomorphism of topological rings as both sides are finite dimensional vector spaces over the complete field K_v and thus have a unique topology induced by the sup-norm (which by uniqueness of norm-topologies on f.d. vector spaces, is the same as the topology on $\prod_i L_i$). Next, the image of the canonical embedding $L \hookrightarrow L \otimes_K K_v$ defined by $\ell \mapsto \ell \otimes 1$ is dense because $K \subseteq L$ is dense in K_v : given any $\ell \otimes x$ in $L \otimes_K K_v$ with $\ell \in L$ and $x \in K_v$, choose $y \in K^\times$ arbitrarily close to x so that $\ell y \otimes 1 = \ell \otimes y$ is an element of the image of L arbitrarily close to $\ell \otimes x$ (and similarly for sums of pure tensors). The image of L is therefore also dense in $\prod_i L_i$ and has dense image under the projections $\prod_i L_i \rightarrow L_i$ and $\prod_i L_i \rightarrow L_i \times L_j$ ($i \neq j$). Now, if $\varphi(L_i) = L_w$ then $L_i \cong L_w$ since L is dense in the complete field L_i , and L_w is the completion of L with respect to the restriction of the absolute value on L_i to L , by the universal property of completions ([4]).

For injectivity of φ , note that if $\varphi(L_i) = \varphi(L_j) = L_w$ for some $i \neq j$ then image of the diagonal embedding $L \rightarrow L_w \times L_w$ is not dense in $L_w \times L_w$ (its closure is isomorphic to $L_w!$), but the image of L is dense in $L_i \times L_j$; a contradiction.

For surjectivity of φ , for each $w|v$ define a continuous homomorphism of finite étale K_v -algebras and topological rings:

$$\varphi_w : L \otimes_K K_v \rightarrow L_w \quad \ell \otimes x \mapsto \ell x.$$

The map φ_w is surjective because its image contains L and is complete, and L_w is the completion of L . It follows that φ_w factors through the projection of $L \otimes_K K_v \cong \prod_i L_i$ on to one of its factors L_i and induces a homeomorphism from L_i to L_w . But then $L_i \cong L_w$ as topological fields, so $\varphi(L_i) = L_w$ and φ is surjective.

Corollary 2.1.6: Places and Irreducible Factors

Let L/K be a finite separable extension of global fields, ν a place of K , and $f \in K[x]$ an irreducible polynomial such that $L \cong K[x]/(f(x))$. Then there is a bijective correspondence between the irreducible factors of f in $K_\nu[x]$ and the places of L lying above ν . In particular, if $f = f_i \cdots f_r$ are the factors of f in $K_\nu[x]$, then:

$$L_{w_i} \cong \frac{K_\nu[x]}{(f_i(x))} \quad 1 \leq i \leq r$$

for appropriate w_i 's.

Proof:
exercise.

Corollary 2.1.7: Classifying Embeddings

Let L/K be a finite separable extension of global fields and ν a place of K . Then given a fixed algebraic closure $\overline{K}_\nu$, there is a bijection

$$\frac{\text{Hom}_K(L, \overline{K}_\nu)}{\text{Gal}_{K_\nu}(\overline{K}_\nu)} \leftrightarrow \{w|\nu\}$$

where the left hand side is the $\text{Gal}_{K_\nu}(\overline{K}_\nu)$ -orbits of K -embeddings of L into $\overline{K}_\nu$, and the right hand side is the places of L above ν .

Proof:

Let $L \cong K(\alpha) = k[x]/(f)$ for some $\alpha \in L$ with minimal polynomial $f \in K[x]$. Then from field theory, there is a bijection between $\text{Hom}_K(L, \overline{K}_\nu)$ and the roots of α_i of f in $\overline{K}_\nu$ that is compatible with the action of $\text{Gal}_{K_\nu}(\overline{K}_\nu)$ on both sets.

Now, if $f = f_1 \cdots f_r$ is the factorization of f in $K_\nu[x]$, each f_i corresponds to an orbit of the galois action on the roots of f , which by corollary 2.1.6 we get a bijective correspondence with the places of L above ν , as we sought to show.

In the special case of $K = \mathbb{Q}$ and $v = \infty$, we have that

$$\frac{\text{Hom}_{\mathbb{Q}}(L, \mathbb{C})}{\text{Gal}_{\mathbb{R}}(\mathbb{C})} \leftrightarrow \{w|\infty\}$$

which gives the Archimedean embedding theorem from [4, chapter 22]

Definition 2.1.8: Real and Complex Place

Let K be a number field. Then the elements of $\text{Hom}_{\mathbb{Q}}(K, \mathbb{R})$ are the *real embeddings* and the elements of $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ whose image does not lie strictly in $\mathbb{R}$ are called the *complex embeddings*

Corollary 2.1.9: Index Given Real and Complex Embeddings

Let K be a number field with r real places and s complex places. Then:

$$[K : \mathbb{Q}] = r + 2s$$

Proof:

This is computation: take $K \cong \mathbb{Q}[x]/(f)$ for some irreducible separable $f \in \mathbb{Q}[x]$ so that:

$$[K : \mathbb{Q}] = \deg f = \#\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$$

Then the action of $\text{Gal}_{\mathbb{R}}(\mathbb{C})$ on $\text{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ as r orbits of size 1 and 2 orbits of size 2, and the rest follows.

Example 2.1.10: Real and Complex Places in Non-trivial Field

Take $K = \mathbb{Q}[x]/(x^3 - 2)$. Then there are three embeddings $K \hookrightarrow \mathbb{C}$ for each root given by:

$$x \mapsto \sqrt[3]{2} \quad x \mapsto \zeta_3 \sqrt[3]{2} \quad x \mapsto \zeta_3^2 \sqrt[3]{2}$$

The first is a real embedding, the other two complex and conjugate to each other. Thus, K has a real place and a complex place, and we get that $[K : \mathbb{Q}] = 1 \cdot 1 + 2 \cdot 1 = 3$.

2.2 Extending Prime Factorization to Places

In this section, we shall continue the analogy between primes and places by showing that they respect many of the similar identities and properties that prime ideals do. We want to show an analogy for residue field, norm, inertia degree and ramification index, the fundamental identity, picard group, and divisors. In this section, we shall use the words prime and place interchangeably, distinguishing them as finite and infinite primes.

Let us first define the residue field for infinite primes. We shall see that

$$\kappa(\mathfrak{p}) = K_{\mathfrak{p}} \quad \mathfrak{p} \mid \infty$$

will be the right notion. Now, for each prime $\mathfrak{p}$ we define the (canonical) homomorphism:

$$\nu_{\mathfrak{p}} : K^{\times} \rightarrow \mathbb{R}$$

where if $\mathfrak{p}$ is finite, it is the usual $\mathfrak{p}$ -adic exponential valuation and $\nu_{\mathfrak{p}}(K^{\times}) = \mathbb{Z}$, and if $\mathfrak{p}$ is infinite then:

$$\nu_{\mathfrak{p}}(a) = -\log(|\tau a|)$$

where $\tau : K \rightarrow \mathbb{C}$ is the embedding which defined $\mathfrak{p}$. Now, for any $\mathfrak{p}/p$ (whether p is finite or infinite, define:

$$f_{\mathfrak{p}} = [\kappa(\mathfrak{p}) : \kappa(p)]$$

Where if $\mathfrak{p} \mid \infty$ we have $f_{\mathfrak{p}} = [K_{\mathfrak{p}} : \mathbb{R}]$ and

$$\mathfrak{N}(\mathfrak{p}) = \begin{cases} p^{f_{\mathfrak{p}}} & \mathfrak{p} \nmid \infty \\ e^{f_{\mathfrak{p}}} & \mathfrak{p} \mid \infty \end{cases}$$

where e here is Euler's number, not the ramification. In fact, we may consider e as an infinite prime number, and the extension $\mathbb{C}/\mathbb{R}$ can be thought of as an unramified extension with inertia degree 2. Next, define the absolute value $|\cdot|_{\mathfrak{p}} : K \rightarrow \mathbb{R}$ via:

$$|a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-\nu_{\mathfrak{p}}(a)}$$

and $|0|_{\mathfrak{p}} = 0$. If $\mathfrak{p}$ is an infinite prime with associated embedding $\tau : K \rightarrow \mathbb{C}$, notice that:

$$|a|_{\mathfrak{p}} = |\tau a| \quad |a|_{\mathfrak{p}} = |\tau a|^2$$

where $\mathfrak{p}$ is real and complex respectively.

Next, if L/K is a finite extension of K , denote the primes $\mathfrak{P}/\mathfrak{p}$ of L the valuations in the class $\mathfrak{P}$ which, when restricted to K , give $\mathfrak{p}$. In the case of an infinite prime $\mathfrak{P}$, we get the inertia degree and ramification index to be:

$$f_{\mathfrak{P}/\mathfrak{p}} = [L_{\mathfrak{P}} : K_{\mathfrak{p}}] \quad e_{\mathfrak{P}/\mathfrak{p}} = 1$$

Thus, we may general the fundamental identity and many other results

Proposition 2.2.1: Fundamental Identity For Places

Let $\mathfrak{P}/\mathfrak{p}$ be an extension of primes (finite or infinite). Then:

1. $\sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}/\mathfrak{p}} f_{\mathfrak{P}/\mathfrak{p}} = \sum_{\mathfrak{P}/\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [L : K]$
2. $\mathfrak{N}(\mathfrak{P}) = \mathfrak{N}(\mathfrak{p})^{f_{\mathfrak{P}/\mathfrak{p}}}$
3. $\nu_{\mathfrak{P}}(a) = e_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{p}}(a)$ (for $a \in K^\times$)
4. $\nu_{\mathfrak{p}}(N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(a)) = f_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{P}}(a)$ (for $a \in L^\times$)
5. $|a|_{\mathfrak{P}} = |N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(a)|_{\mathfrak{p}}$ (for $a \in L$)

Proof:

exercise

2.2.1 Galois Theory of Global Fields

Next, let us take a moment to extend the theory of places when L/K is a galois extension of global field. Then we have a galois group $\text{Gal}_K(L)$ acting on the set of places ω of L via $\omega \mapsto \sigma(w)$ where $\sigma(w)$ is the equivalence classes of absolute valued given by

$$\|\alpha\|_{\sigma(w)} = \|\sigma(\alpha)\|_w$$

This generalizes the case from the finite prime case. We can generalize the decomposition group:

Definition 2.2.2: Decomposition Group For Places

Let L/K be a galois extension of global fields and let w be a place of L . Then the *decomposition group* of w is the stabilizer:

$$D_w = \{\sigma \in \text{Gal}_K(L) \mid \sigma(w) = w\}$$

We shall see that in this case we can get infinite primes that ramify. By letting $L \cong \mathbb{Q}[x]/(f)$, we get a one-to-one correspondence between the embeddings of L into $\mathbb{C}$ with the root of $f \in \mathbb{C}$. Then each embedding of L into $\mathbb{C}$ restricts to an embedding of K into $\mathbb{C}$. This map induces a map that sends each infinite place w of L to an infinite place v of K that extends to w . Now, this map may send a complex place to a real place, which happens when a pair of distinct complex conjugate embeddings of L restrict to the same embedding of K (implying it must be a real embedding). In this case, we say that v (and w) is *ramified* in the extension L/K , and the ramification index is $e_v = 2$ (and $e_v = 1$

otherwise). In this case, we shall also have $f_\nu = 1$ for $\nu \mid \infty$, and $g_\nu = \#\{w \mid \nu\}$. With all of this, we may generalize the fundamental identity to:

$$e_\nu f_\nu g_\nu = [L : K]$$

and more generally:

Definition 2.2.3: Ramification Index For Global Infinite Primes

Let L/K be a galois extension of number fields L/K . Then:

$$e_0(L/K) = \prod_{\nu \nmid \infty} e_\nu \quad e_\infty(L/K) = \prod_{\nu \mid \infty} e_\nu$$

$$e(L/K) = e_0(L/K)e_\infty(L/K)$$

Now as L/K is galois, it induces a transitive action on the embeddings and their corresponding places. As $L \cong k[x]/(g)$, each embedding of K into $\mathbb{C}$ gives rise to $[L : K]$ distinct embeddings of L into $\mathbb{C}$ that extend it, one for each root of g in $\mathbb{C}$. Thus, for each infinite place ν of K , the galois group acts transitively on $\{w \mid \nu\}$, and so either all of the places $w \mid \nu$ are ramified or none are, which is determined by the decomposition group for places (namely, whether the decomposition group is order 1 or 2).

Let us conclude by generalizing Dirichlet's unit theorem, which is named after *Herband* as it shall have important results linking to Herband's Quotient we reach section ??:

Theorem 2.2.4: Herband Unit Theorem

Let L/K be a galois extension of number field. Let $w_1, \dots, w_r$ be the real places of L and $w_{r+1}, \dots, w_{r+s}$ be the complex places of L . Then there exists $\epsilon_1, \dots, \epsilon_{r+s} \in \mathcal{O}_L^\times$ such that

1. $\sigma(\epsilon_i) = \epsilon_j$ if and only if $\sigma(w_i) = w_j$ for all $\sigma \in \text{Gal}_K(L)$
2. $\epsilon_1, \dots, \epsilon_{r+s}$ generate a finite index subgroup of $\mathcal{O}_L^\times$
3. $\epsilon_1 \epsilon_2 \cdots \epsilon_{r+s} = 1$ and every relation among the ϵ_i is generated by this one.

Proof:

This proof is straight from Sutherland's notes. I need some more time to internalize it before typing it out myself. It is here for ease of reference

1. Pick $\epsilon_1, \dots, \epsilon_{r+s} \in \mathcal{O}_L^\times$ such that $\|\epsilon_i\|_{w_j} < 1$ for $i \neq j$; the existence of such ϵ_i follows from the strong approximation theorem that we will prove in the next lecture; the product formula then implies $\|\epsilon_i\|_{w_i} > 1$. Now let $\alpha_i := \prod_{\sigma \in D_{w_i}} \sigma(\epsilon_i) \in \mathcal{O}_L^\times$. We have $\|\alpha_i\|_{w_i} = \prod_{\sigma \in D_{w_i}} \|\epsilon_i\|_{w_i} > 1$ and $\|\alpha_i\|_{w_j} = \prod_{\sigma \in D_{w_i}} \|\epsilon_i\|_{\sigma(w_j)} < 1$, since $\sigma \in D_{w_i}$ fixes w_i and permutes the w_j with $j \neq i$. Each α_i is fixed by D_{w_i} .

Let $G := \text{Gal}(L/K)$. For $i = 1, \dots, r+s$, let $r(i) := \min\{j : \sigma(w_j) = w_i \text{ for some } \sigma \in G\}$, so that $w_{r(i)}$ is a distinguished representative of the G -orbit of w_i . For $i = 1, \dots, r+s$ let $\beta_i := \sigma(\alpha_{r(i)})$, where σ is any element of G such that $\sigma(w_{r(i)}) = w_i$. The value of $\sigma(\alpha_{r(i)})$ does not depend on the choice of σ because $\sigma_1(w_{r(i)}) = \sigma_2(w_{r(i)})$ if and only if

$\sigma_2^{-1}\sigma_1 \in D_{w_{r(i)}}$ and $\alpha_{r(i)}$ is fixed by $D_{w_{r(i)}}$. The β_i then satisfy (i).

2. The β_i also satisfy (2): a product $\gamma_j := \prod_{i \neq j} \beta_i^{n_i}$ cannot be trivial because $\|\gamma_j\|_{w_j} < 1$; in particular, $\beta_1, \dots, \beta_{r+s-1}$ generate a subgroup of $\mathcal{O}_L^\times$ isomorphic to $\mathbb{Z}^{r+s-1}$ which necessarily has finite index in $\mathcal{O}_L^\times \simeq \mathbb{Z}^{r+s-1} \times \mu_L$ (see Theorem 15.12). But we must have $\prod_i \beta_i^{n_i} = 1$ for some tuple $(n_1, \dots, n_{r+s}) \in \mathbb{Z}^{r+s}$ (with $n_i = n_j$ whenever w_i and w_j lie in the same G -orbit, since every $\sigma \in G$ fixes 1). The set of such tuples spans a rank-1 submodule of $\mathbb{Z}^{r+s}$ from which we choose a generator $(n_1, \dots, n_{r+s})$ (by inverting some β_i if necessary, we can make all the n_i positive if we wish). Then $\varepsilon_i := \beta_i^{n_i}$ satisfy (1), (2), (3) as desired.

2.2.2 Picard Group for Places

Let us extend fractional ideals of K by taking into account infinite primes

Definition 2.2.5: Replete Ideal

Let K be a field. Then a *replete ideal* of K is an element of the group:

$$J(\overline{\mathcal{O}}) = J(\mathcal{O}) \times \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^\times$$

For notational consistency, for any infinite prime $\mathfrak{p}$ and any real number $\nu \in \mathbb{R}$, let:

$$\mathfrak{p}^\nu := e^\nu \in \mathbb{R}_+^\times$$

Next, given a system of real numbers $\nu_{\mathfrak{p}}, \mathfrak{p}|\infty$, let $\prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}}$ denote the vector:

$$\prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}} = (\dots, e^{\nu_{\mathfrak{p}}}, \dots) \in \prod_{\mathfrak{p}|\infty} \mathbb{R}_+^\times$$

and *not* the product of the quantities $e^{\nu_{\mathfrak{p}}} \in \mathbb{R}$. Then every replete ideal $\mathfrak{a} \in J(\overline{\mathcal{O}})$ admits a unique factorization:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{\nu_{\mathfrak{p}}} \times \prod_{\mathfrak{p}|\infty} \mathfrak{p}^{\nu_{\mathfrak{p}}}$$

we shall denote the finite portion $\mathfrak{a}_f$ and the infinite portion $\mathfrak{a}_\infty$ so that

$$\mathfrak{a} = \mathfrak{a}_f \times \mathfrak{a}_\infty$$

Then $\mathfrak{a}_f$ is a fractional ideal and $\mathfrak{a}_\infty$ is an element of $\prod_{\mathfrak{p}|\infty} \mathbb{R}_+^\times$. Viewing $\mathfrak{a}_f$ and $\mathfrak{a}_\infty$ as elements of

$$\mathfrak{a}_f \times \prod_{\mathfrak{p}|\infty} 1 \quad (1) \times \mathfrak{a}_\infty$$

we can think of all the elements of $J(\overline{\mathcal{O}})$ as

$$\mathfrak{a} = \mathfrak{a}_f \cdot \mathfrak{a}_\infty$$

Now, to each $a \in K^\times$, we can associated to it a *replete principal ideal*:

$$|a| = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{u_{\mathfrak{p}}}(a)}$$

This forms a subgroup $P(\overline{\mathcal{O}}) \subseteq J(\overline{\mathcal{O}})$ and hence we can define the *replete picard group* or *replete class group*:

Definition 2.2.6: Replete Picard Group

$$\text{Pic}(\overline{\mathcal{O}}) = \frac{J(\overline{\mathcal{O}})}{P(\overline{\mathcal{O}})}$$

Next,

Definition 2.2.7: Absolute Norm For Replete Ideals

The *absolute norm* of a replete ideal $\mathfrak{a}$ is defined to be the positive real number:

$$\mathfrak{N}(\mathfrak{a}) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}$$

The absolute norm is multiplicative and induces a surjective homomorphism:

$$\mathfrak{N} : J(\overline{\mathcal{O}}) \rightarrow \mathbb{R}_+^\times$$

By using a generalization of proposition 1.1.10, the replete norm of a principal ideal a is equal to 1

$$\mathfrak{N}(a) = 1$$

Hence, we in fact get a surjective homomorphism on the replete Picard group

$$\mathfrak{N} : \text{Pic}(\overline{\mathcal{O}}) \rightarrow \mathbb{R}_+^\times$$

There is still a bit more in Neukirch, p.187. I gotta leave it alone for now to finish other work, but must come back

2.3 *Haar Measure and Product Formula

A natural property that the real and complex numbers have is the existence of a *translation invariant Lebesgue measure* (unique up to scaling; or unique if the unit interval/square must have measure 1). With the places introduced, it is natural to ask whether a measure can be defined on $\overline{K}_\nu$ and if these measure have some similarities to the real and euclidean one. As $\overline{K}_\nu$ are locally compact abelian groups, there is always a unique *Haar measure* defined on them. The build-up for this measure would require a course in analysis (see [3] for the full details), so the following is a summary of the important concepts

2.3.1 Sutherland vs. these notes n Sutherland, this is the beginning of what is the “lattice” section in [4, chapter 22]. I am covering this here because the Haar measure of local fields cam up in my research, but it is not strictly necessary for the rest of these notes.

Definition 2.3.2: Radon Measure and Haar Measure

Let X be a locally compact Hausdorff space with Borel measure μ . Then μ is a *Radon measure* if for every Borel set S :

1. $\mu(S) < \infty$ if S is compact
2. $\mu(S) = \inf \{\mu(U) \mid S \subseteq U, U \text{ is open}\}$
3. $\mu(S) = \sup \{\mu(C) \mid C \subseteq S, C \text{ is compact}\}$

If X is furthermore a group, a Radon measure is a (left/right) *Haar measure* if it is (left/right) *translation invariant*:

$$\mu(S) = \mu(x + S) \quad [\text{for right } \mu(S + x)]$$

If X is abelian, the left/right measures are identical and the measure is simply called a *Haar measure*.

Theorem 2.3.3: Existence and Uniqueness of Haar Measure

Let G be a locally compact group. Then there exists a Haar measure μ on G . If μ' is another Haar measure, then there exists a $\lambda \in \mathbb{R}_{>0}$ such that $\mu'(S) = \lambda\mu(S)$.

Proof:

see [3]

Example 2.3.4: Haar Measure On Real Space

The standard Haar measure on $\mathbb{R}^n$ is the familiar:

$$\mu \left(\prod_i (a_i, b_i) \right) = \prod_i |b_i - a_i|$$

which is also the unique Haar measure where the cube has measure 1.

Let us apply this to local fields. By proposition 1.2.3, every local field K is locally compact, and the corresponding valuation ring A is compact, in particular $A = B_1(0)$. Given a Haar measure μ on K , it shall always be normalized so that $\mu(A) = 1$; with this condition this makes the μ the unique Haar measure on K .

Proposition 2.3.5: Haar Measure On Local Fields

Let K be a local field with discrete valuation v , residue field k , and absolute value

$$|\cdot|_v := (\#k)^{-v(\cdot)},$$

and let μ be a Haar measure on K . For every $x \in K$ and measurable set $S \subseteq K$ we have

$$\mu(xS) = |x|_v \mu(S).$$

Moreover, the absolute value $|\cdot|_v$ is the unique absolute value compatible with the topology on K for which this is true.

Proof:

This is straight from Sutherland!

Let A be the valuation ring of K with maximal ideal $\mathfrak{p}$. The proposition clearly holds for $x = 0$, so let $x \neq 0$. The map $\varphi_x : y \mapsto xy$ is an automorphism of the additive group of K , and it follows that the composition $\mu_x = \mu \circ \varphi_x$ is a Haar measure on K , hence a multiple of μ , say $\mu_x = \lambda_x \mu$, for some $\lambda_x \in \mathbb{R}_{>0}$. Define the function $\chi : K^\times \rightarrow \mathbb{R}_{>0}$ by $\chi(x) := \lambda_x = \mu_x(A)/\mu(A)$. Then $\mu_x = \chi(x)\mu$, and for all $x, y \in K^\times$ we have

$$\chi(xy) = \frac{\mu_{xy}(A)}{\mu(A)} = \frac{\mu_x(yA)}{\mu(A)} = \frac{\chi(x)\mu_y(A)}{\mu(A)} = \frac{\chi(x)\chi(y)\mu(A)}{\mu(A)} = \chi(x)\chi(y).$$

Thus χ is multiplicative, and we claim that in fact $\chi(x) = |x|_v$ for all $x \in K^\times$. Since both χ and $|\cdot|_v$ are multiplicative, it suffices to consider $x \in A - \{0\}$. For any such x , the ideal xA is equal to $\mathfrak{p}^{v(x)}$, since A is a DVR. The residue field $k := A/\mathfrak{p}$ is finite, hence A/xA is also finite; indeed it is a k -vector space of dimension $v(x)$ and has cardinality $[A : xA] = (\#k)^{v(x)}$. Writing A as a finite disjoint union of cosets of xA , we have

$$\mu(A) = [A : xA]\mu(xA) = (\#k)^{v(x)}\chi(x)\mu(A),$$

and therefore $\chi(x) = (\#k)^{-v(x)} = |x|_v$ as claimed. It follows that

$$\mu(xS) = \mu_x(S) = \chi(x)\mu(S) = |x|_v \mu(S),$$

for all $x \in K$ and measurable $S \subseteq K$. To prove uniqueness, if $|\cdot|$ is an absolute value on K that induces the same topology as $|\cdot|_v$ then for some $0 < c < 1$ we have $|x| = |x|_v^c$ for all $x \in K^\times$. Let us fix $x \in K^\times$ with $|x|_v \neq 1$ (take any x with $v(x) \neq 0$). If $|\cdot|$ also satisfies $\mu(xS) = |x|\mu(S)$ then

$$\frac{\mu(xA)}{\mu(A)} = |x| = |x|_v^c = \left(\frac{\mu(xA)}{\mu(A)} \right)^c,$$

which implies $c = 1$, meaning that $|\cdot|$ and $|\cdot|_v$ are the same absolute value.

Exercise 2.3.1

1. Let K be a number field with s complex places. Show that the sign of the absolute discriminant Δ_K is $(-1)^s$.

2.4 *Arakelov Compactification

As we have seen, the infinite places behave analytically like boundary points to the finite primes. In standard algebraic geometry, the affine scheme $\text{Spec}(\mathcal{O}_K)$ is not proper (compact). To geometrically compactify this space, we must formally append the infinite places to create a space often denoted $\widehat{\text{Spec}}(\mathcal{O}_K)$.

However, as the “local ring” at an infinite place is a closed bounded interval (which is not closed under addition), we cannot simply glue a new affine scheme using prime ideals. Instead, Arakelov geometry compactifies the space by altering the category of divisors and line bundles, gluing analytic metrics onto the algebraic data.

Definition 2.4.1: Arakelov Divisor

Let K be a number field. An *Arakelov divisor* is a formal sum:

$$\widehat{D} = \sum_{\mathfrak{p} \nmid \infty} n_{\mathfrak{p}}(\mathfrak{p}) + \sum_{\nu \mid \infty} x_{\nu}(\nu)$$

where $n_{\mathfrak{p}} \in \mathbb{Z}$ and $x_{\nu} \in \mathbb{R}$. The group of Arakelov divisors is denoted $\widehat{\text{Div}}(K)$.

Notice that the coefficients for the infinite places are allowed to be real numbers, reflecting their continuous, Archimedean nature. This is the additive counterpart to the replete ideals defined in section 2.2.

Definition 2.4.2: Principal Arakelov Divisor

For any $a \in K^{\times}$, its *principal Arakelov divisor* is defined as:

$$\widehat{(a)} = \sum_{\mathfrak{p} \nmid \infty} \nu_{\mathfrak{p}}(a)(\mathfrak{p}) - \sum_{\nu \mid \infty} \log |a|_{\nu}(\nu)$$

The negative sign on the infinite places is a standard convention so that the product formula manifests as a degree-zero condition, exactly as it does for meromorphic functions on a compact Riemann surface (where the number of zeros equals the number of poles).

Definition 2.4.3: Arakelov Degree

The *degree* of an Arakelov divisor $\widehat{D}$ is the homomorphism $\deg : \widehat{\text{Div}}(K) \rightarrow \mathbb{R}$ given by:

$$\deg(\widehat{D}) = \sum_{\mathfrak{p} \nmid \infty} n_{\mathfrak{p}} \log \mathfrak{N}(\mathfrak{p}) + \sum_{\nu \mid \infty} x_{\nu}$$

Theorem 2.4.4: Arakelov Degree of Principal Divisors

Let $a \in K^{\times}$. Then $\deg(\widehat{(a)}) = 0$.

Proof:

This is an immediate consequence of the Product Formula. We have:

$$\deg(\widehat{(a)}) = \sum_{\mathfrak{p} \nmid \infty} \nu_{\mathfrak{p}}(a) \log \mathfrak{N}(\mathfrak{p}) - \sum_{\nu \mid \infty} \log |a|_{\nu}$$

By the definition of the normalized absolute values from section 2.2, exponentiating this sum yields $\prod_{\nu} |a|_{\nu} = 1$. Taking the logarithm of both sides proves the sum is 0.

This theorem shows that by adding the infinite places, $\overline{\text{Spec}(\mathcal{O}_K)}$ behaves exactly like a projective curve. Finally, at the level of sheaves, this compactification is realized by equipping algebraic line bundles with analytic data at the boundary.

Definition 2.4.5: Hermitian (Arakelov) Line Bundle

An *Arakelov line bundle* $\overline{\mathcal{L}}$ on $\overline{\text{Spec}(\mathcal{O}_K)}$ is a pair $(\mathcal{L}, (\|\cdot\|_{\nu})_{\nu \mid \infty})$ where $\mathcal{L}$ is an invertible sheaf on $\text{Spec}(\mathcal{O}_K)$, and for each infinite place ν , $\|\cdot\|_{\nu}$ is a Hermitian metric on the complex vector space $\mathcal{L} \otimes_{\nu} \mathbb{C}$.

Thus, rather than constructing a topological stalk at ∞ , we compactify the line bundle by providing a notion of “size” for its sections at the infinite boundaries.

The machinery of Arakelov divisors and Hermitian line bundles is not just a formal trick; it allows us to import the powerful global theorems of algebraic curves into number theory. For a projective curve X , the group of divisors modulo principal divisors forms the Picard group $\text{Pic}(X)$.

Definition 2.4.6: Arakelov Class Group

The *Arakelov class group* (or arithmetic Picard group) $\widehat{\text{Pic}}(\mathcal{O}_K)$ is defined as the quotient group:

$$\widehat{\text{Pic}}(\mathcal{O}_K) = \widehat{\text{Div}}(K) / \{\widehat{(a)} \mid a \in K^{\times}\}$$

Because $\overline{\text{Spec}(\mathcal{O}_K)}$ is now properly compactified, $\widehat{\text{Pic}}(\mathcal{O}_K)$ behaves like the Picard group of a compact Riemann surface. Its structure elegantly combines the ideal class group of K (from the finite primes) with the unit group $\mathcal{O}_K^{\times}$ (regulated by the infinite metrics). Furthermore, this framework provides the exact geometric setting required to state and prove the Arithmetic Riemann-Roch Theorem, which computes the “dimension” (measured via the volume of Euclidean lattices) of the space of global sections of $\overline{\mathcal{L}}$.

A Remark on $\mathbb{F}_1$ and Purely Algebraic Compactification

While Arakelov geometry provides a rigorous and beautiful hybrid of algebra and complex analysis, the necessity of gluing analytic metrics leaves a lingering geometric dissatisfaction. As noted at the beginning of this section, the “local ring” at an infinite place (such as the closed interval $[-1, 1]$ for the standard Euclidean valuation) is a multiplicatively closed monoid, but it fails to be a commutative ring because it is not closed under addition (e.g., $1 + 1 = 2 \notin [-1, 1]$).

Because Grothendieck's scheme theory and specifically the Proj functor used to algebraically compactify spaces strictly requires commutative rings, the infinite places are fundamentally locked out of classical algebraic geometry.

This strict requirement is the primary motivation behind the highly active research program of $\mathbb{F}_1$ -*geometry* (geometry over the field with one element). In this speculative framework, the foundational algebraic objects are relaxed from rings to monoids. If $\mathbb{Z}$ can be successfully recast as an algebra over the deeper base of $\mathbb{F}_1$, the monoid $[-1, 1]$ would become a perfectly valid local algebra. Under such a theory, the analytic workarounds of Arakelov geometry would no longer be necessary, and $\text{Spec}(\overline{\mathcal{O}_K})$ would emerge naturally as a literal, purely algebraic compactification.

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